BUILDING CONSTRUCTION

Module 01:
MASONRY CONSTRUCTION

BRICK MASONRY

- When bricks are laid in mortar in a proper systematic manner, they form a homogeneous mass, which can withstand forces without disintegration. This mass of the structure, so made by the use of bricks is called "**Brick Masonry**" or simply "**Brick work**".
- Bricks are of uniform size and shape, light in weight, durable, fire resistant, have high resale value, low maintenance cost and are easily available in plain areas.
- Brick Masonry is commonly used for construction of ordinary as well as important buildings in plain areas now-a-days.

SOME IMPORTANT TERMS USED IN BRICK MASONRY

(1) BRICK

- An artificial structural element in the form of a rectangular block of clay is called a "Brick ". Bricks can be manufactured of any required shape and size.

TYPE OF BRICKS	GRADE	PROPERTIES	USES
1ST CLASS BRICK.	A	WELL BURNT,SOUND,SHARP EDGES,PROER SHAPE AND SIZE.	FACE-WORKED STRUCTURE.
2ND CLASS BRICK.	B	IRREGULAR IN SHAPE AND SIZE,BLUNT EDGES.	ORDINARY STRUCTURES, BRICK-WORK, UNDER PLASTERS.
3RD CLASS BRICK.	C	UNDER BURNT BRICKS,SOFT AND LIGHT IN COLOUR.	UNIMPORTANT AND TEMPORARY STRUCTURES,AT HEAVY RAINFALL PLACES.
4TH CLASS BRICK. (ZHAMA BRICK)	D	DEFORMED SHAPE AND SIZE,OVER BURNT.	AGGREGRATES FOR CONCRETE IN FOUNDATION,FLOORS, ROADS, ETC.

SOME IMPORTANT TERMS USED IN BRICK MASONRY

(2) FROG

- The depression provided in the face of a brick is called a "**Frog**".
- It is provided in the brick to achieve the following purposes:

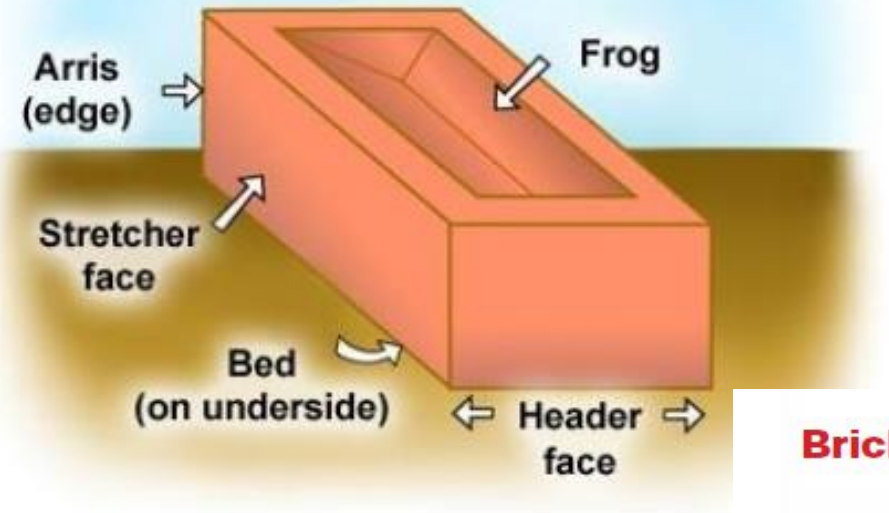
  (a) To form a key of mortar in between any two adjacent courses of brick work, so as to increase the lateral strength of the structure.

  (b) To reduce the weight of the bricks, so that the bricks can be laid with convenience.

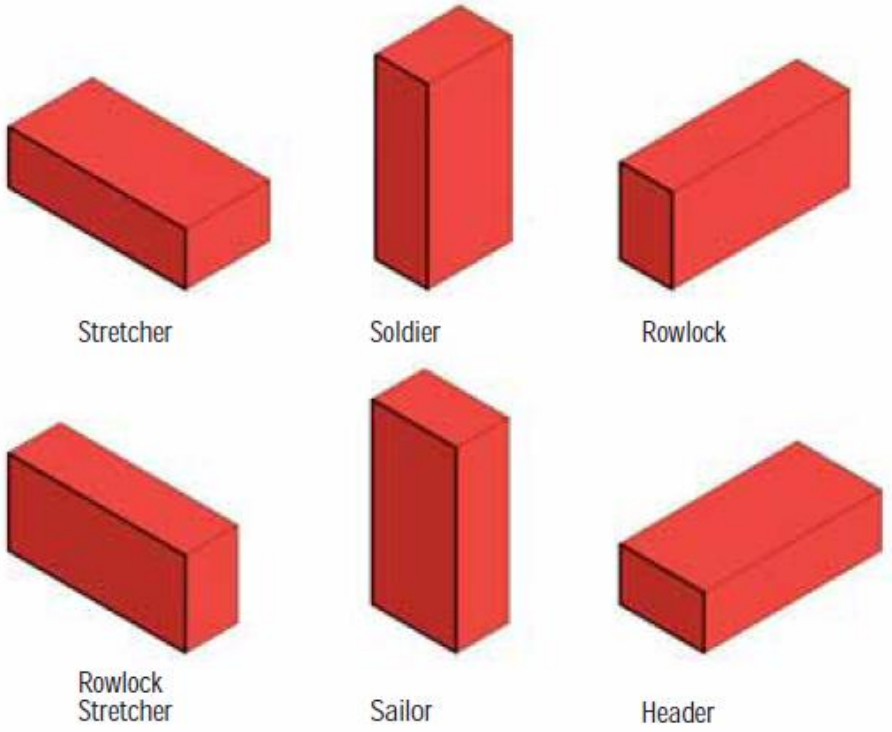
  (c) To provide a place for putting the impression of trade-mark or the year of manufacturing of the bricks

(3) POSITION OF BRICKS

- (a) The position of brick, when laid with its Frog upward in the horizontal plane, is termed as "**Brick on bed**".
- (b) The position of the brick when laid on its side "9 in x 3 in", with frog in the vertical plane is called "**Brick on edge**".
- (c) The position of brick when laid on its side "4 1/2 in x 3 in", with frog in the vertical plane is called " **Brick on end**".



Brick Positions in a Wall



(4) COURSE

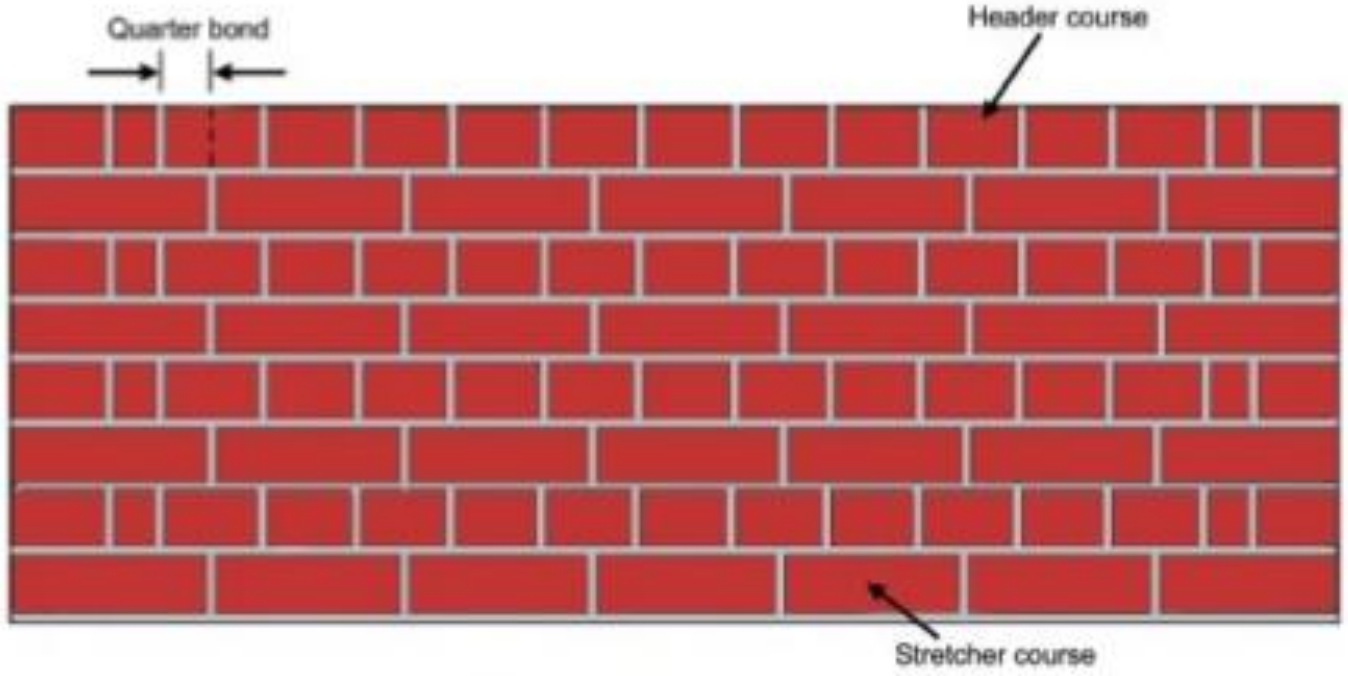
Each horizontal layer of bricks laid in mortar in a brick work is called a "**course**".

(5) STRETCHER

Brick, laid with its length horizontal and parallel with the face of the wall or other masonry member is called a "**Stretcher**" and a course, in which, all the bricks are laid as Stretchers is called a "Stretching course" or "Stretcher course".

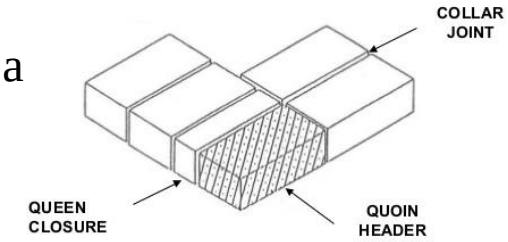
(6) HEADER

A brick laid, so that only its end shows on the face of a wall is called a **"Header"** and a course, in which all the bricks are laid as headers, is known as "**Heading Course**" or **"Header course".**



(7) QUOIN

The external corner of the wall is called a "**Quoin**".





(8) QUOIN BRICK

The brick, which forms the external corner of a wall is known as " **Quoin brick**".

(9) QUOIN HEADER

A corner header, in the face of wall, which is a stretcher in the side wall is known as "**Quoin header**".

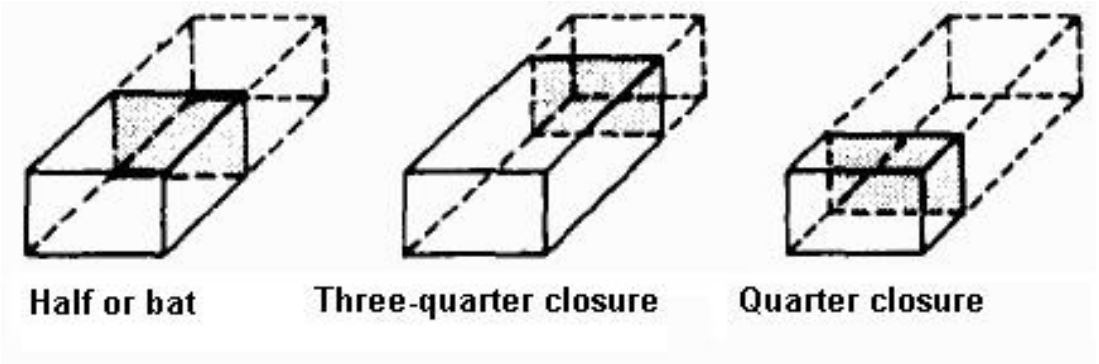
(10) QUOIN STRETCHER

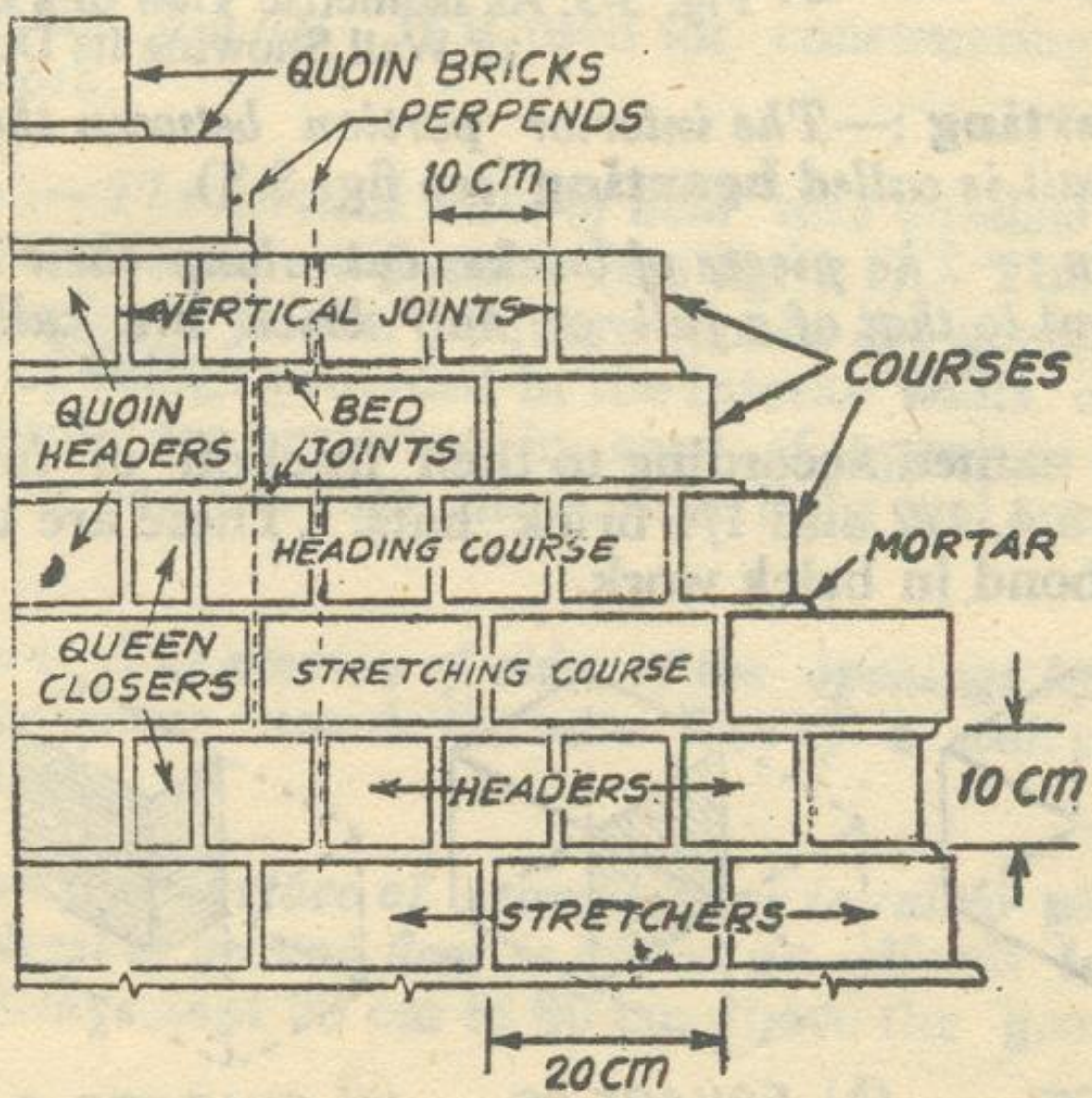
A corner stretcher in the face of a wall, which is header in the side wall is known as "**Quoin stretcher**".

(11) BRICK BATS

The pieces of bricks, cut long their length and having width equivalent to that of a full or half brick are called "**Brick bats**".

Some common Brick Bats are shown below:





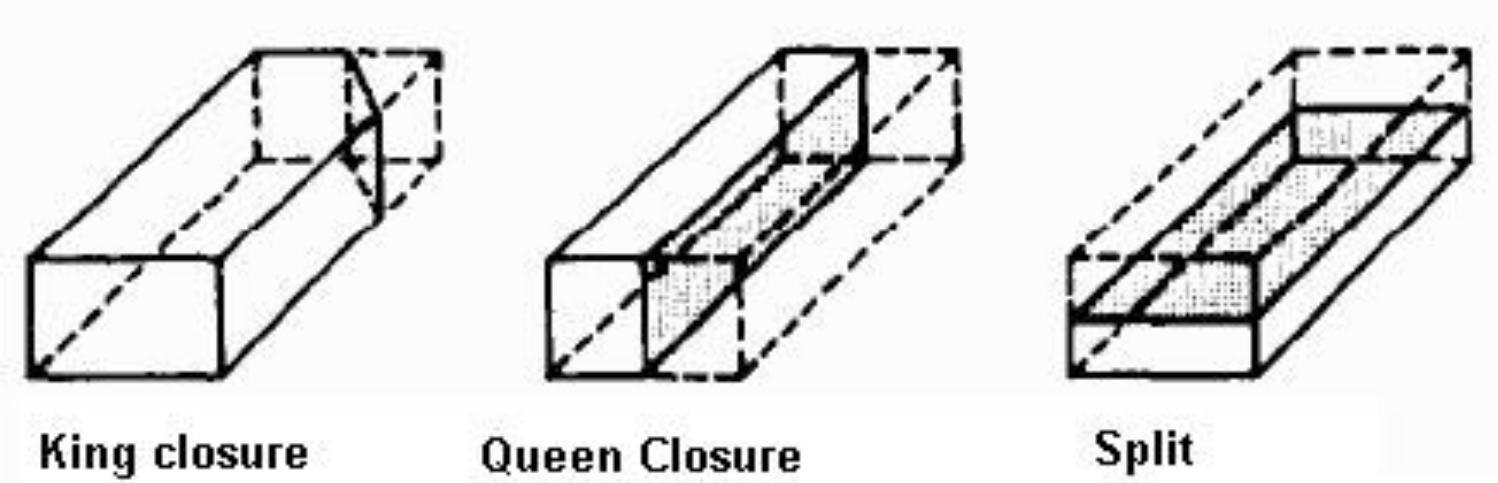
Elevation of the portion of a Wall Showing its Different Details

(12) QUEEN CLOSER

- **Queen closer** is a brick, which is half as wide as full brick and is made by cutting a whole brick lengthwise into two portions.
- These are generally used next to the Quoin header for creating bonds in brickwork.

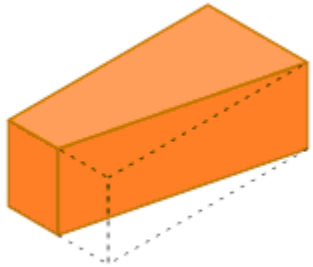
(13) KING CLOSER

- A brick, whose one diagonal piece is cut off one corner by a vertical plane passing through the center of one end to the center of one side.
- It is actually 7/8 of a full brick but is usually called a 3/4 brick



(14) BEVELED CLOSER

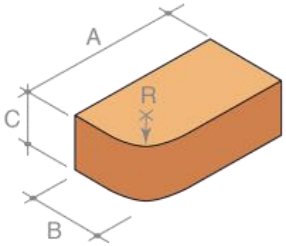
- A brick cut longitudinally along a vertical plane, starting at the middle of one end to the far corner.
- One quarter of the brick is cut off in this way.



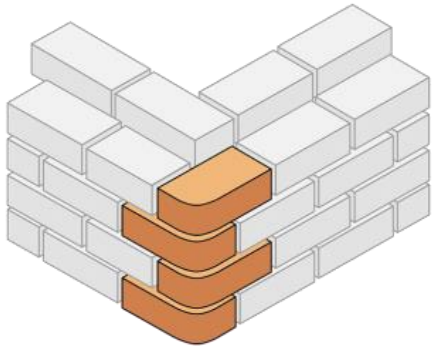
Bevelled Closer

(15) BULL NOSE BRICK

- A brick with rounded corners is called a "**Bull Nose Brick**"

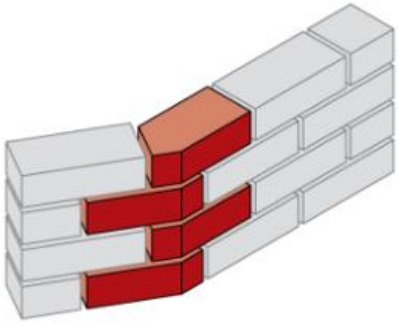


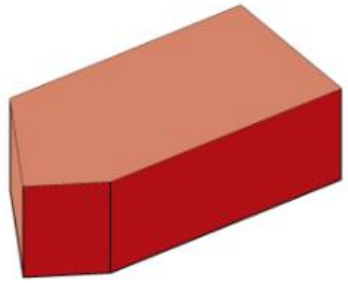
Left hand shown



(16) SQUINT BRICKS

➢These bricks are used to construct acute (>90 degree) or obtuse (< 90 degree) corners in brick masonry.
➢These are special forms of bricks.





(17) JAMB

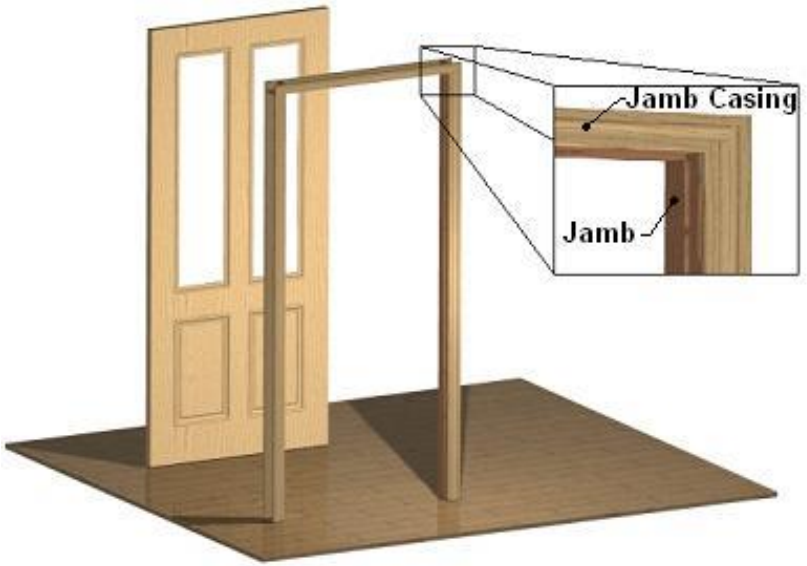
The vertical sides of door or window openings provided in a wall are known as **"Jambs".**

(18) REVEALS

The part of the Jamb opening , which is exposed between a door or window frame and the face or back of a wall is known as " **Reveal".**

(19) SILL

The horizontal part (either of timber, concrete, stone, metal, etc) at the bottom of a door or window, supporting the vertical members of the frame is known as " **Sill** " and its height window base from the floor level is known as **" Sill level ".**



(20) MORTAR

- The paste obtained by mixing a binding material and a fine aggregate in suitable proportions in addition to water is known as "**Mortar**".
- Cement and Lime are used as binding materials and Sand, Surkhi, Cinder, etc. are used as fine aggregates.
- The mortars are named according to the type of binding material used in their preparation such as, cement mortar, lime mortar, etc.
- The mortar prepared from simple earth is known as "Mud Mortar".

BONDS

Bond is the arrangement of bricks or stones in each course, so as to ensure the greatest possible interlocking and to avoid the continuity of vertical joints in two successive courses, both on the face and in the body of a wall.

OBJECTIVES OF BONDS

A bond is provided to achieve the following objectives:

(a) The primary objective of providing a bond is to break the continuity of the vertical joints in the successive courses both in the length and thickness of masonry structure. As a result, the structure will act as a bounded mass and its load will be transmitted uniformly to the foundations.

(b) To ensure longitudinal and lateral strength of the structure.

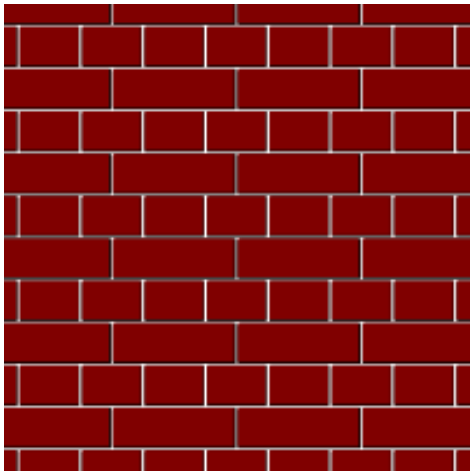
(c) To provide pleasing appearance by laying bricks symmetrically.

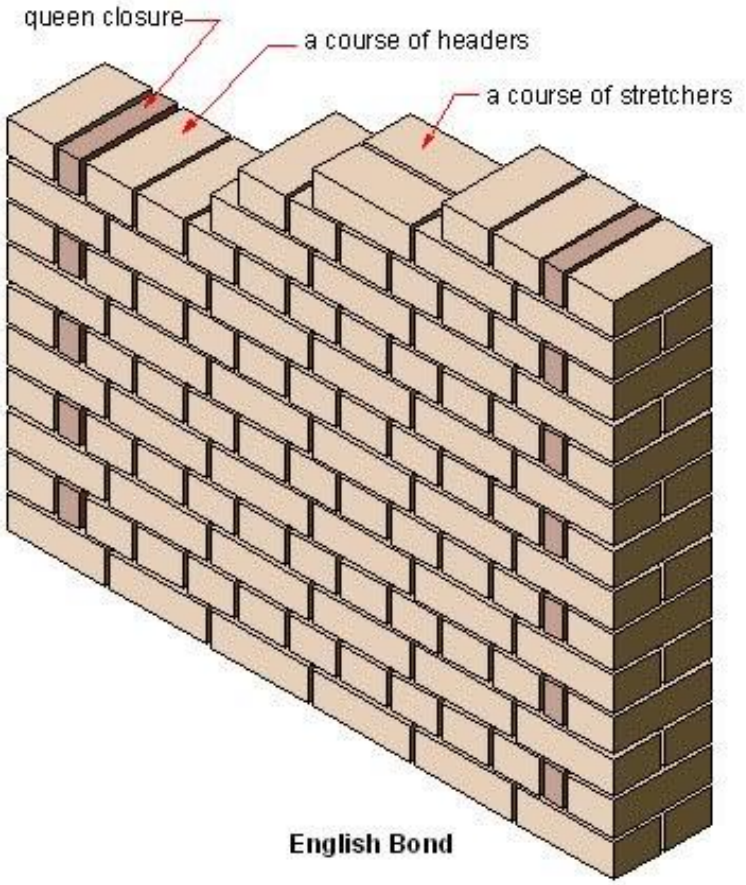
(d) To do masonry work quickly by engaging more masons on a job at a time.

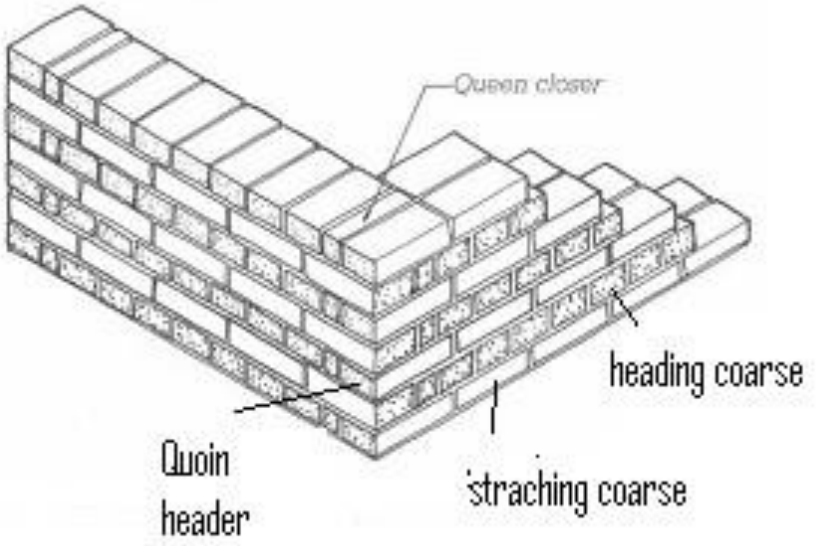
TYPES OF BONDS

(1) ENGLISH BOND

The bond, in which headers and stretchers are laid in alternate courses, is called **"English bond".**







ENGLISH BOND

The following are the salient features of English bond:

(**i**) Headers and stretchers are laid in alternate courses.

(**ii**) In each heading course, a queen closer is placed next to quoin header and the remaining bricks are laid as headers.

(**iii**) Every alternate header in a course comes centrally over the joint between two stretchers in the course below, giving an approximate lap of 2 ¼ in

(**iv**) The same course will show headers or stretchers on face and back, if the thickness of the wall is an even multiple of half bricks (e.g. 9 in, 18 in, 27 in, etc.)

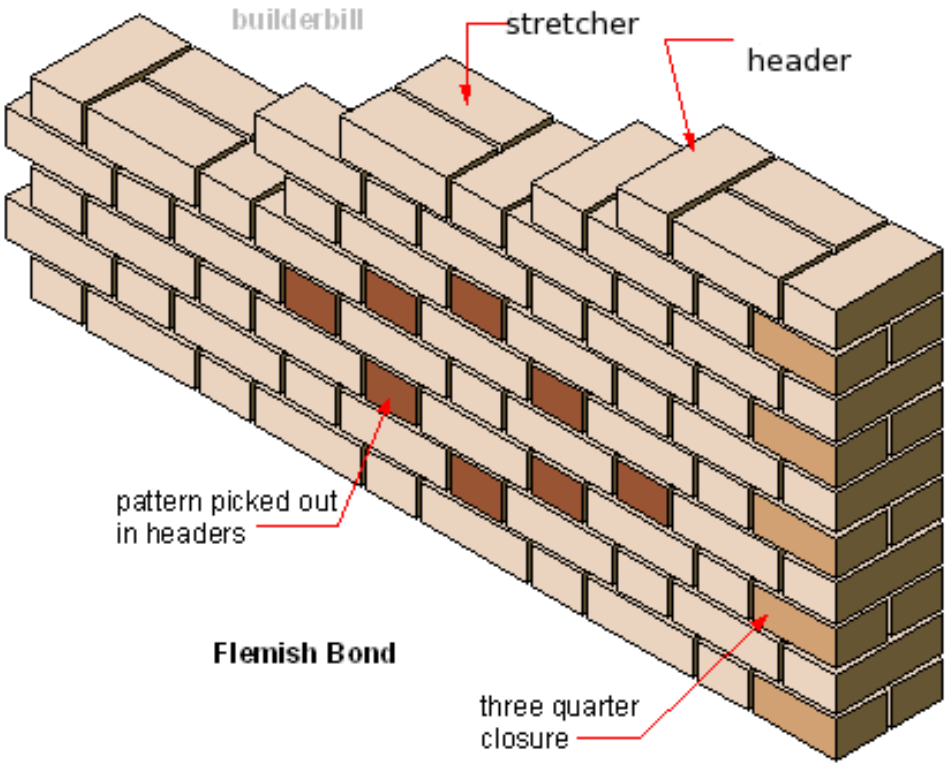
(**v**) The same course will show headers on the face and stretchers on the back and vice versa, if the thickness of the wall is an odd multiple of half brick. (13 1/2 in , 22 1/2 in , etc)

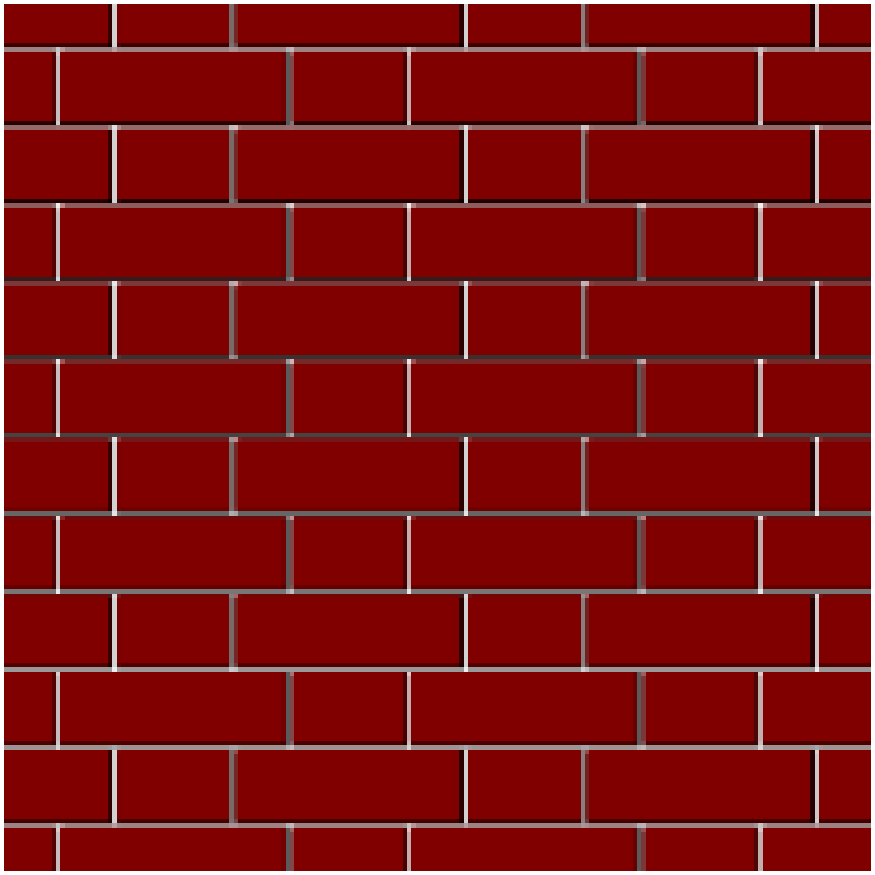
(**vi**) The middle portion of the thicker walls consists entirely of headers.

(**vii**) Every transverse joint is continuous from face to face.

(2) FLEMISH BOND:

The bond, in which headers and stretchers are laid alternately in the same course, is called **"Flemish bond".**





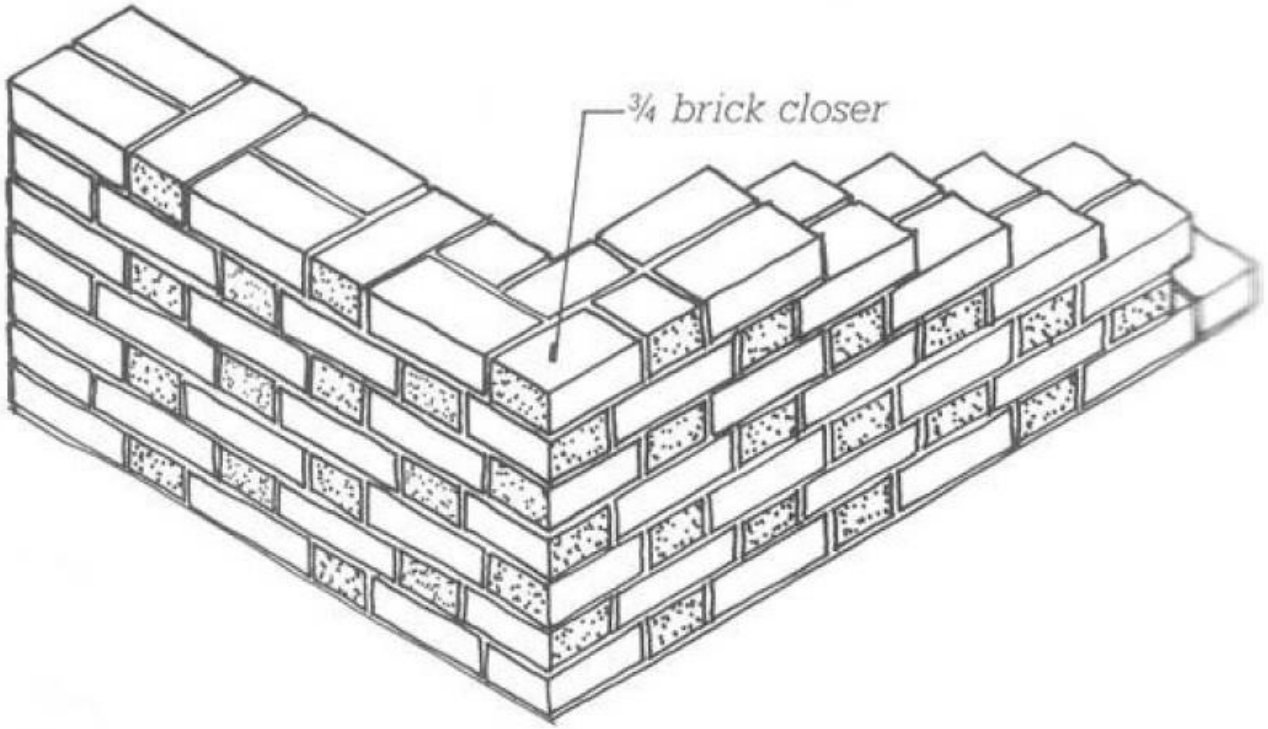
The following are the salient features of Flemish bond:

(**i**) Headers and stretchers are laid alternately in the same course.

(**ii**) Every header in each course lies centrally over every stretcher of the underlying course.

(**iii**) In every alternate course a queen closer is placed next to quoin header, so as to provide a lap of approximately 2 1/4 in.

(**iv**) Brick bats are to be used in walls having thickness equal to an odd multiple of half brick.



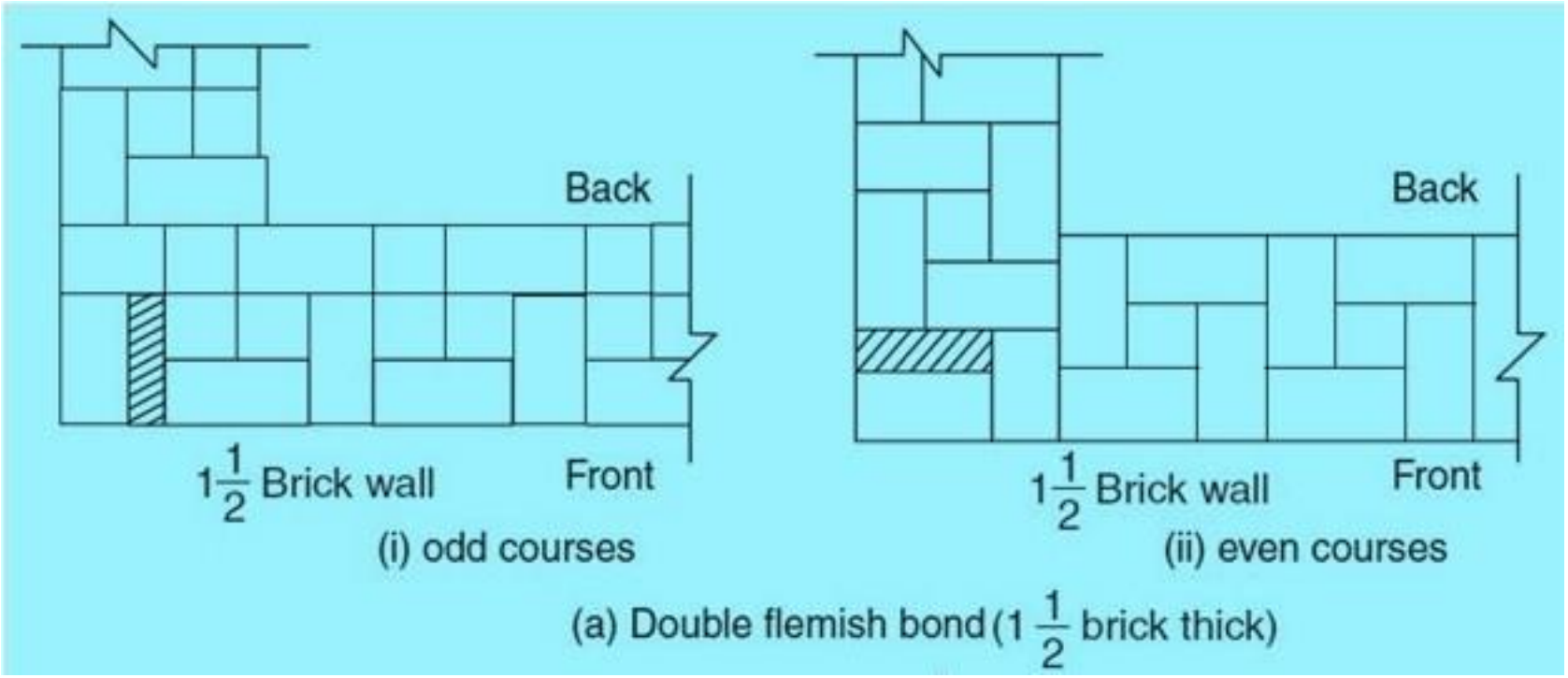
Comparison of English Bond and Flemish bond

Sr No	*English Bonds*	*Flemish bond*
1	This bond consists of headers and stretchers laid in alternative courses.	This bond consists of headers and stretchers laid alternatively in each course.
2	It is strongest of all the bonds.	It is less strong for walls having thickness more than 13 ½ inches.
3	It provides rough appearance especially for one brick thick walls.	It provides good appearance for all thickness of walls.
4	There are no noticeable continuous vertical joints in the structure built in this bond.	There are partly continuous vertical joints in the structure built in this bond.
5	Much attention is not required in providing this bond.	Special attention is required in providing this bond.
6	Progress of work is more.	Progress of work is less.
7	It is costly because the use of brick bats is not allowed.	It is economical because brick bats are allowed for forming this bind.

TYPES OF FLEMISH BONDS

(a) DOUBLE FLEMISH BOND:

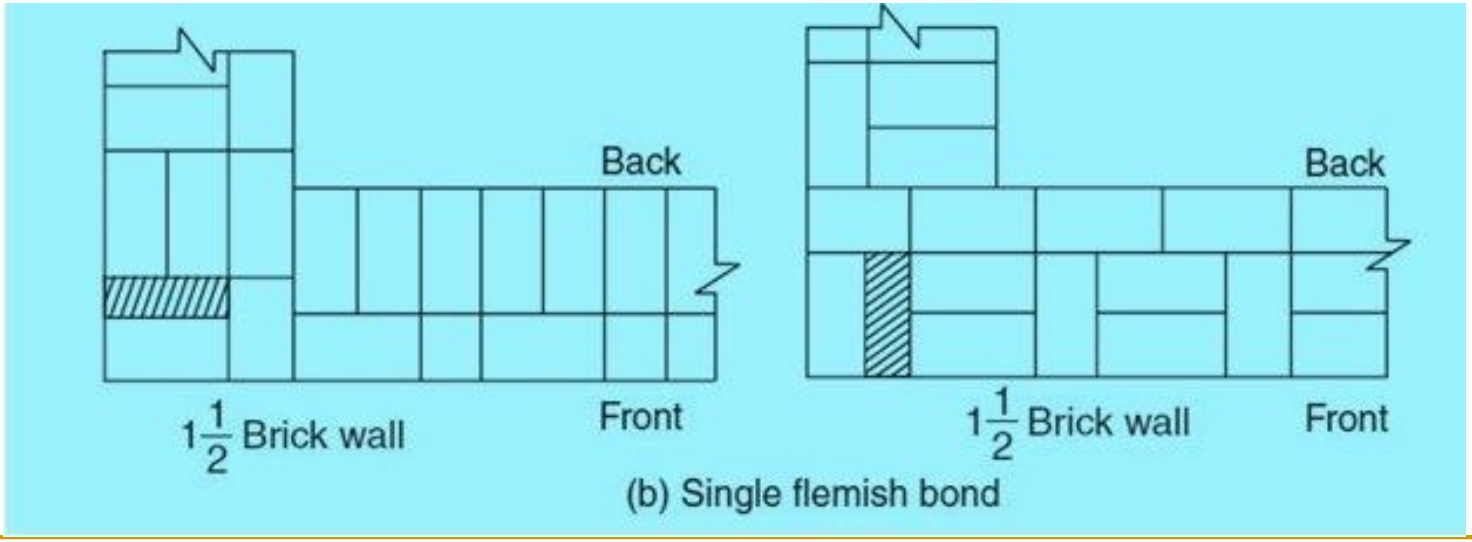
The bond in which headers and stretchers are laid alternately in each course, both in the face and back of the wall, is called **Double Flemish Bond**.



(a) Double flemish bond ($1\frac{1}{2}$ brick thick)

(b) SINGLE FLEMISH BOND

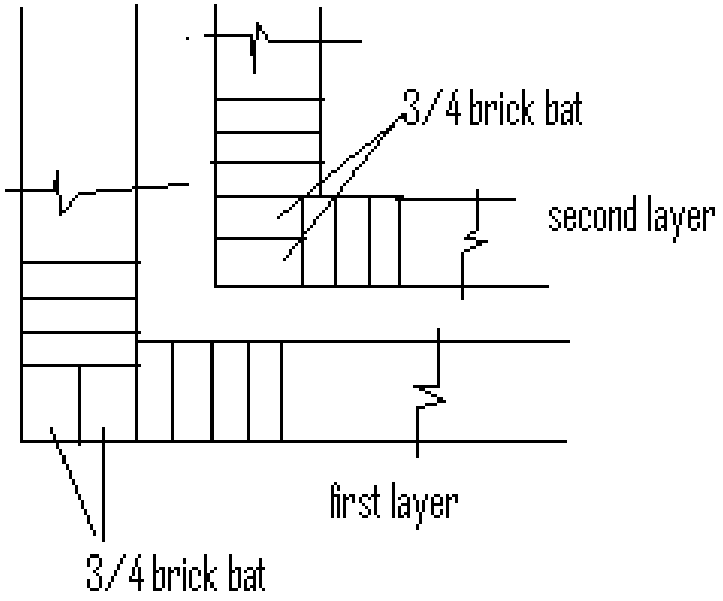
- The bond provided in a wall with Flemish bond in facing and English bond in backing is called **"Single Flemish bond" or "Cross bond".**
- This bond combines the advantages of both English and Flemish bonds and simultaneously eliminates their disadvantages.
- This bond is recommended where costly bricks are specified for facing in order to provide good appearance to the wall. Also, it can be made more economical by using cheap quality of bricks on the back of wall.
- On the other hand, it weakens the overall strength of the wall because of maximum use of brick and existence of continuous vertical joints. Also, it can not be provided in walls having thickness less than 13 ½ in.

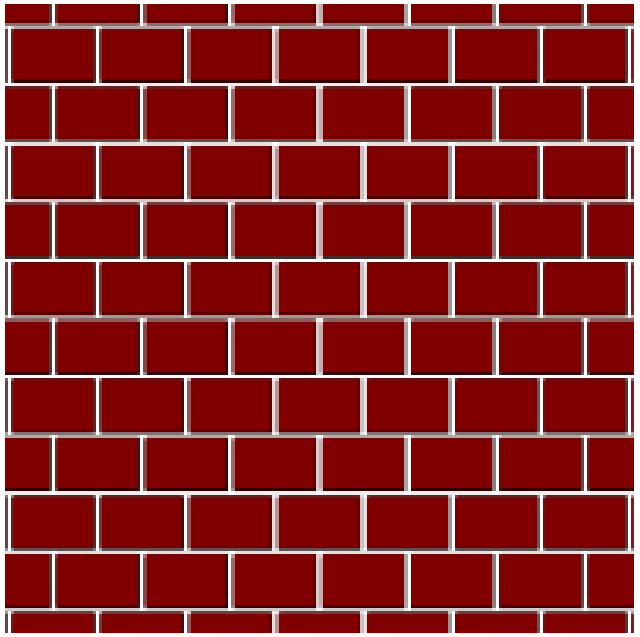


(b) Single flemish bond

(3) HEADING BOND

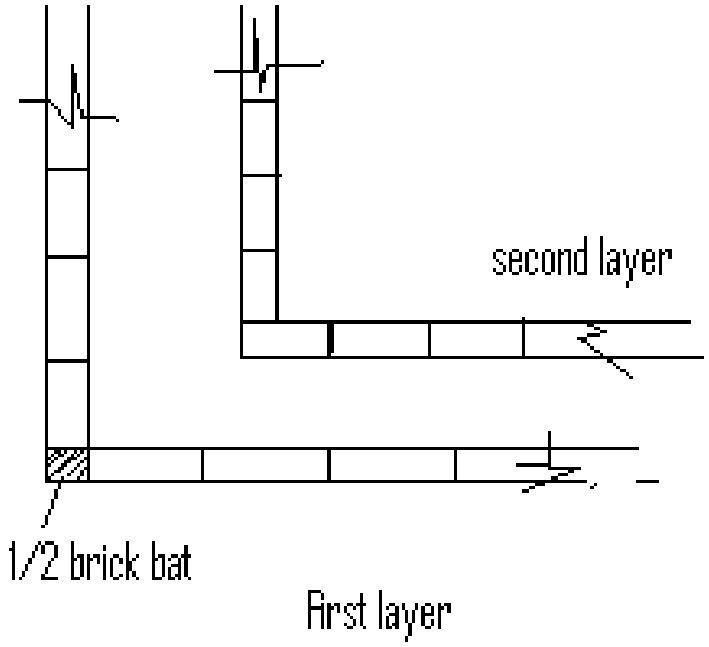
- The bond in which all the bricks are laid as headers in every course of a wall is called **"Heading bond".**
- 3/4 bats are laid as quoin bricks in the alternate courses to break the continuity of vertical joints, which increases the transverse strength but weakens the longitudinal strength of the wall.
- This bond is commonly used for constructing steining of wells, footings of walls and columns, corbels, cornices, etc.

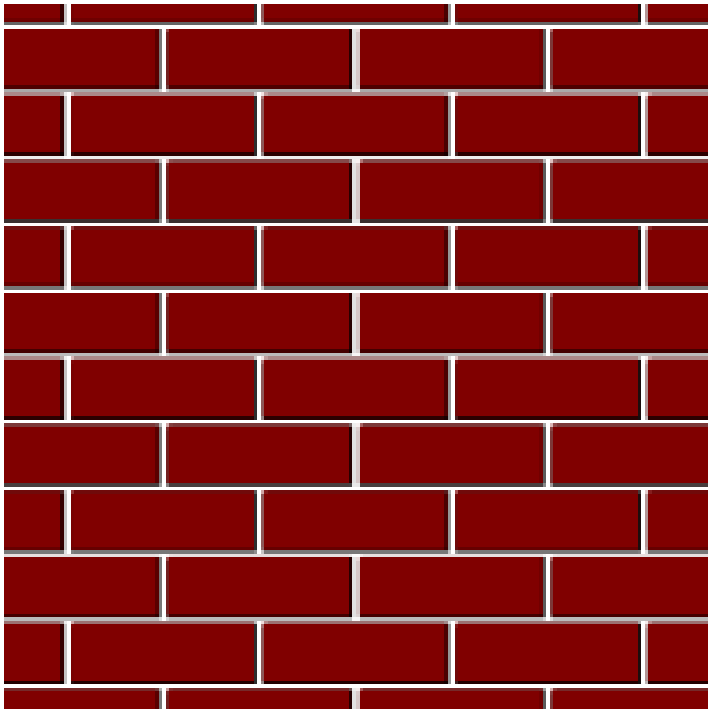




(4) STRETCHING BOND

- The bond in which all the bricks are laid as stretchers in every course is called "**Stretching bond**”.
- This bond is provided for constructing 4 ½ in thick partition walls.
- Only used for walls of half brick thickness(partition wall) , this is the only practical bond which can be used on a wall of this thickness.to break the vertical continuity ½ brick bat is provided in alternating courses



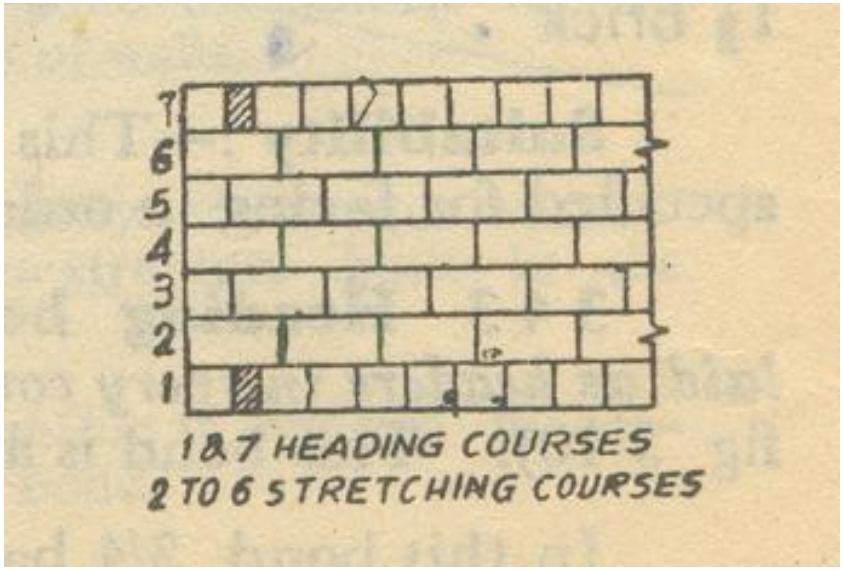


(5) GARDEN WALL BOND

This bond is used for constructing one brick thick garden walls, boundary walls, and other walls such as outer leaves of cavity walls to provide good appearance.

(a) ENGLISH GARDEN WALL BOND

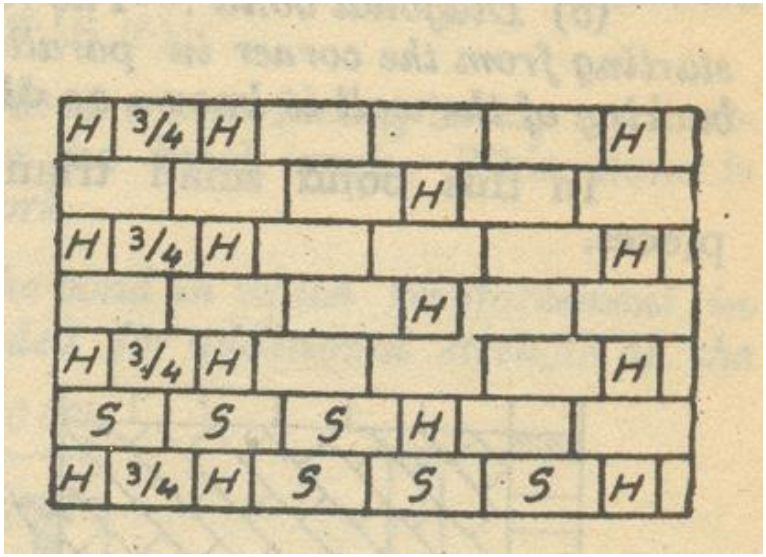
The garden wall bond in which a heading course is provided after 3 or 5 stretching courses is called "**English Garden Wall Bond**".



(b) FLEMISH GARDEN WALL BOND

In this bond a header is provided after 3 or 5 stretches in each course.

This bond is also known as "**Sussex or Scotch Bond**".

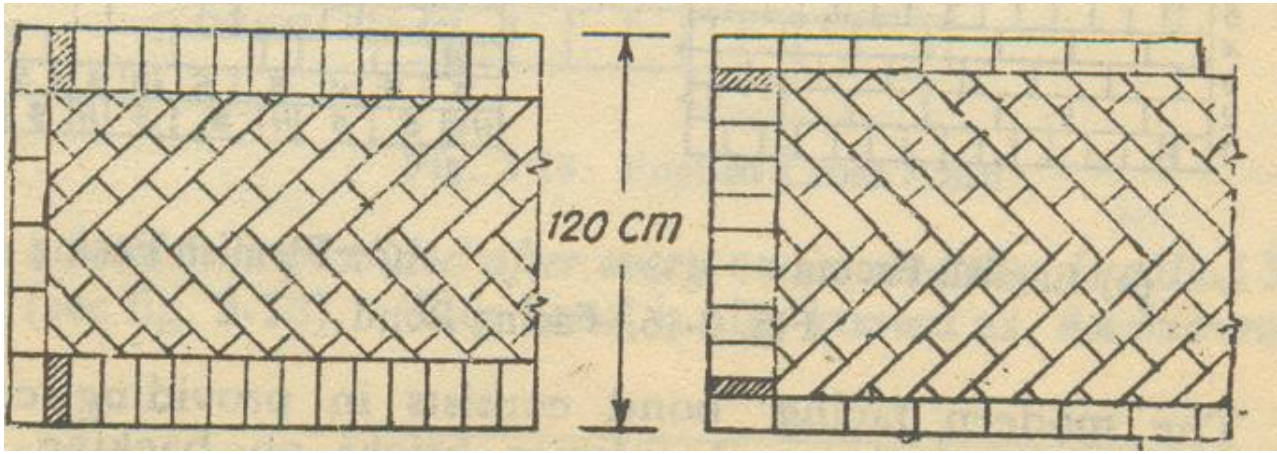


(6) RAKING BOND

- The bond in which all the bricks are laid at an angle other than 90^0 to the facing and backing of the wall is known as **"Raking bond".**
- This bond is used for doing inner filling of walls at suitable intervals to improve their longitudinal strength.
- The angle of rake between any two adjacent courses should be 90 degree to attain maximum transverse strength of the wall.
- This bond can also be used as paving in case of brick floors, 4 ½ in thick.

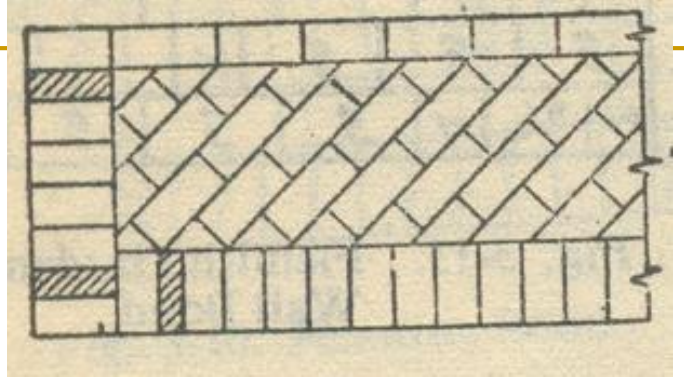
(a) HERRING BONE BOND

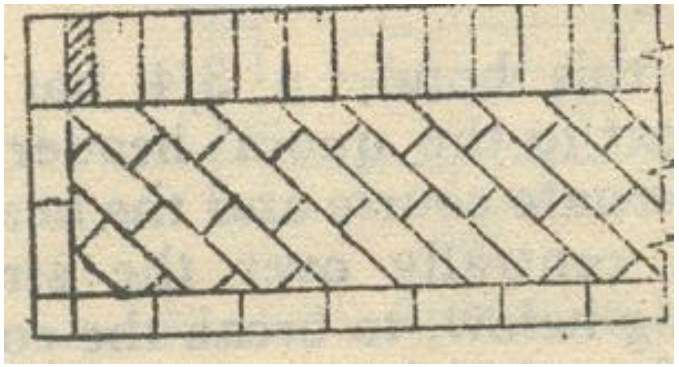
The raking bond in which bricks are laid at an angle of 45 degree , starting at the central line and proceeding towards the facing and backing of the wall, is called **"Herring Bone Bond"**



(b) DIAGONAL BOND

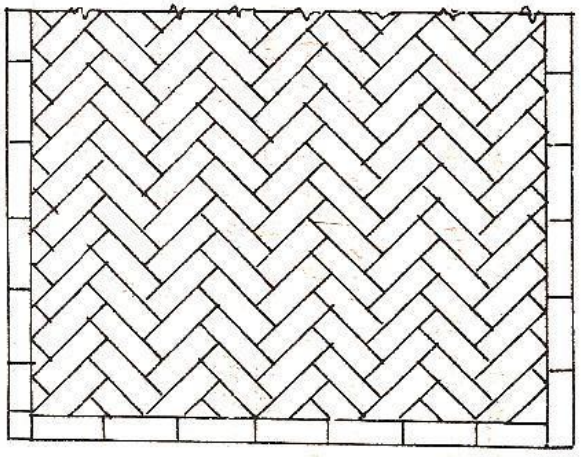
The raking bond in which bricks are laid starting from the corner in parallel rows inclined to the facing and backing of the wall is known as **"Diagonal bond".**





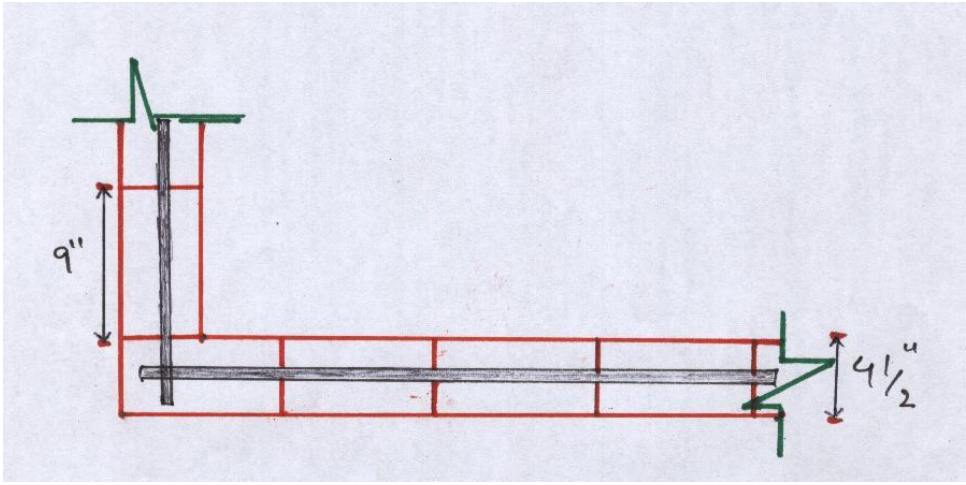
(10)ZIG-ZAG BOND

- Bond is similar to herring-bone bond except that bricks are laid in zigzag fashion .
- Generally used for making ornamental panels in brick flooring.



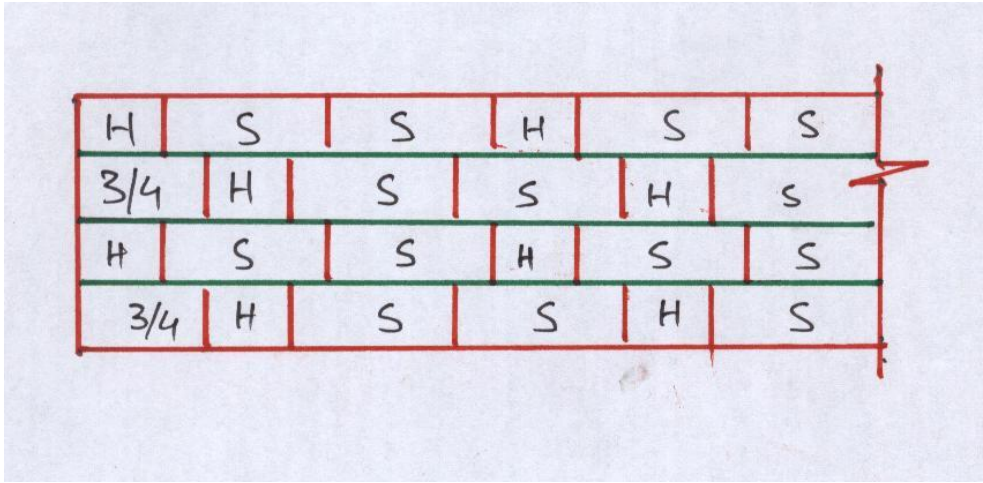
(7) HOOP IRON BOND

- The bond in which, after every fourth or fifth course of masonry, reinforcement in the form of longitudinal ties is provided for additional strength of the wall, is called " **Hoop Iron Bond".**
- This bond is provided for constructing 4 1/2 in thick partition walls



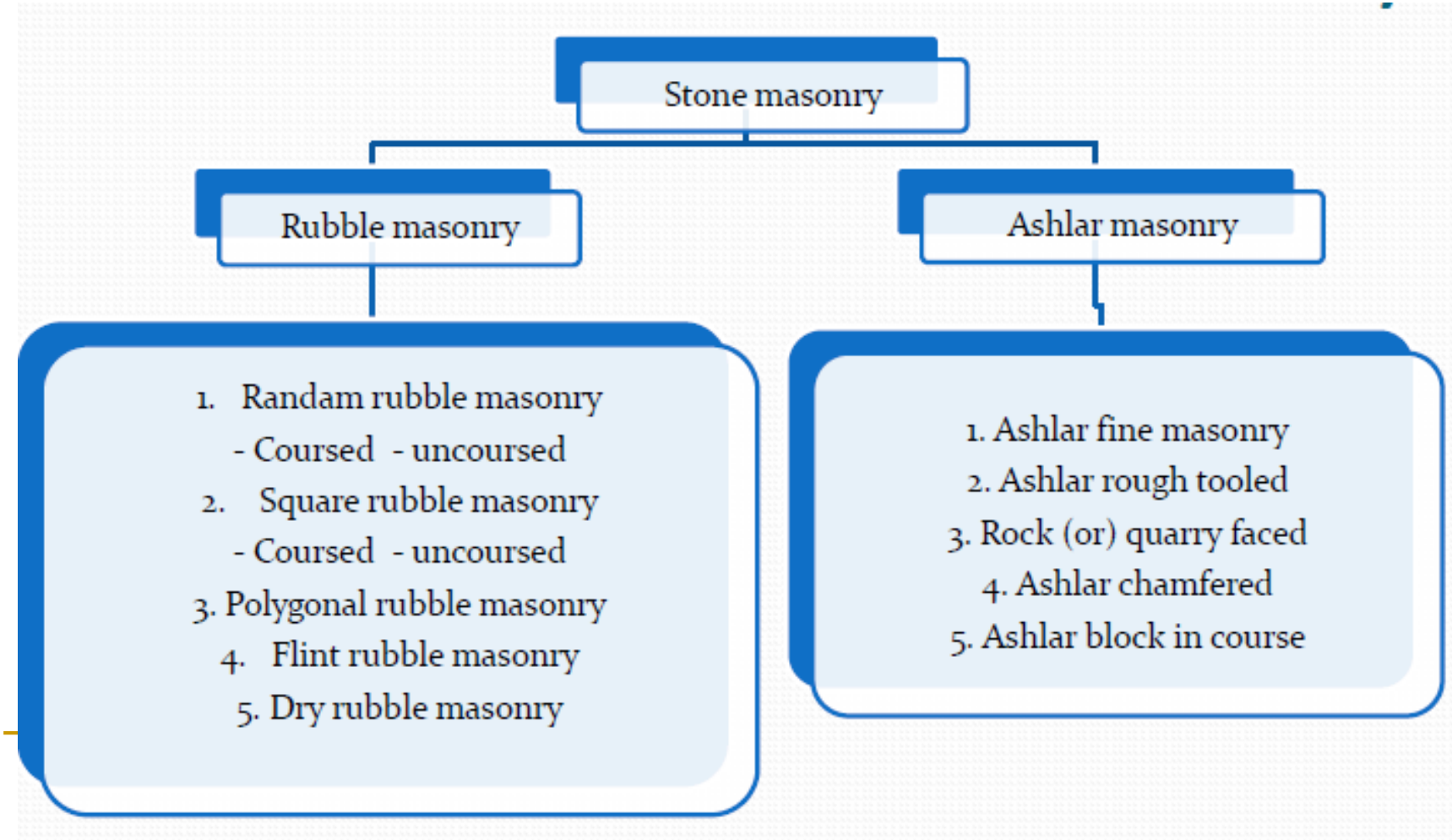
(8) MONK BOND

- This bond in which two stretchers and one header are laid alternately in each course is called **"Monk bond**".
- This bond is used in the construction of boundary walls.



STONE MASONRY

Rock, that is removed from its natural site and generally, cut or dressed and then finished for building purposes, is called **"Stone"** and the art of building the structure with stones as constructional units is called **"Stone Masonry"**.



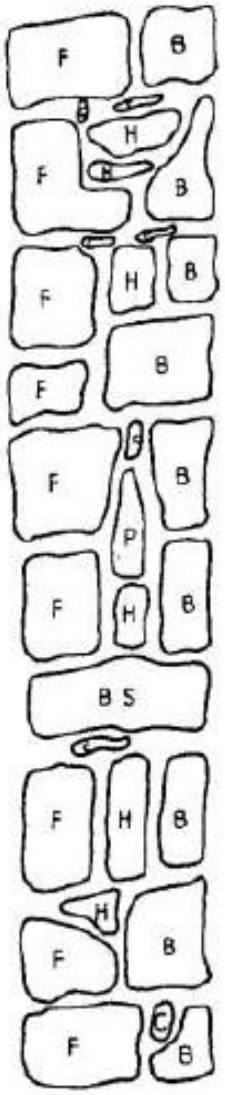
RUBBLE MASONRY

- The stone masonry in which either undressed or roughly dressed stones are laid is called "**Rubble masonry**".
- In this masonry, the joints of mortar are not of uniform thickness.



(a) RANDOM RUBBLE MASONRY

- The rubble masonry in which either undressed or hammer dressed stones are used is called "**Random Rubble Masonry**".
- The strength of this masonry depends upon the bond between the stones.
- The bond should be sound both transversely and longitudinally.
- Transverse bond is obtained by the liberal use of "Bonders" and "Troughs"

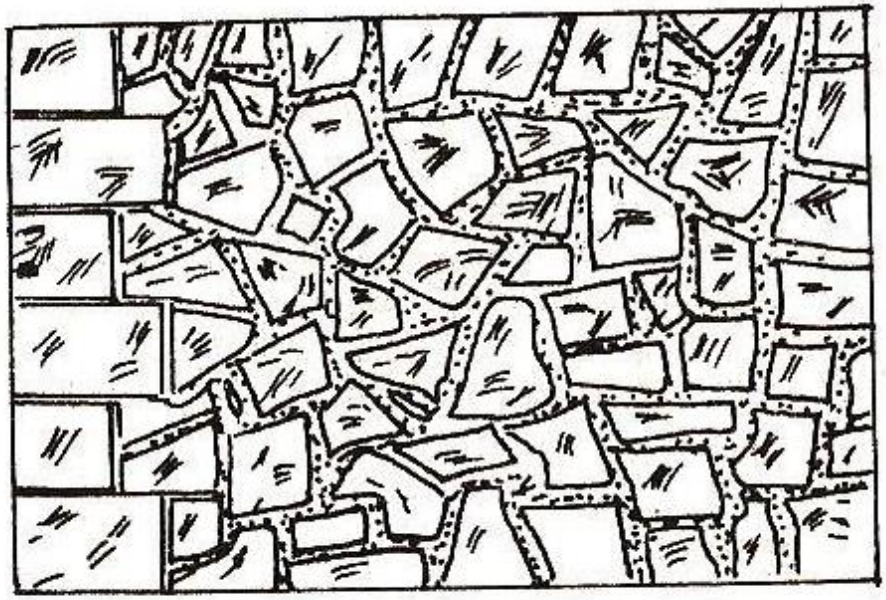


B - BACKING STONE
H - HEARTING STONE
P - PLUM STONE
BS - BOND STONE
C - CHIP STONE
F - FACING STONE

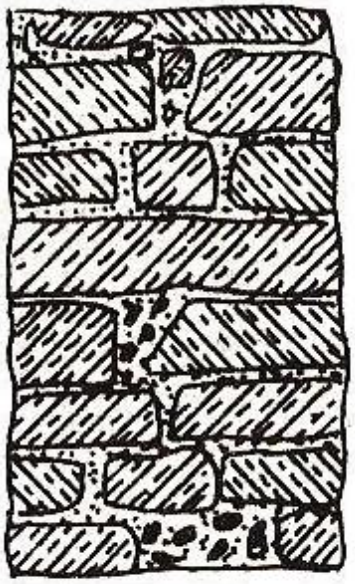


(i) UNCOURSED RANDOM RUBBLE MASONRY

- The random rubble masonry, in which all the stones are laid without forming courses, is known as **"Uncoursed Random Rubble masonry"**.
- This masonry is the cheapest and roughest type of masonry.
- The stones to be used in this masonry are of different sizes and shapes.
- Larger stones are used at corners and at jambs to increase their strength.
- In general, stones are laid with their longer axis roughly horizontal and along the length of the wall.
- This type of masonry is used for constructing walls of low height in case of ordinary buildings.



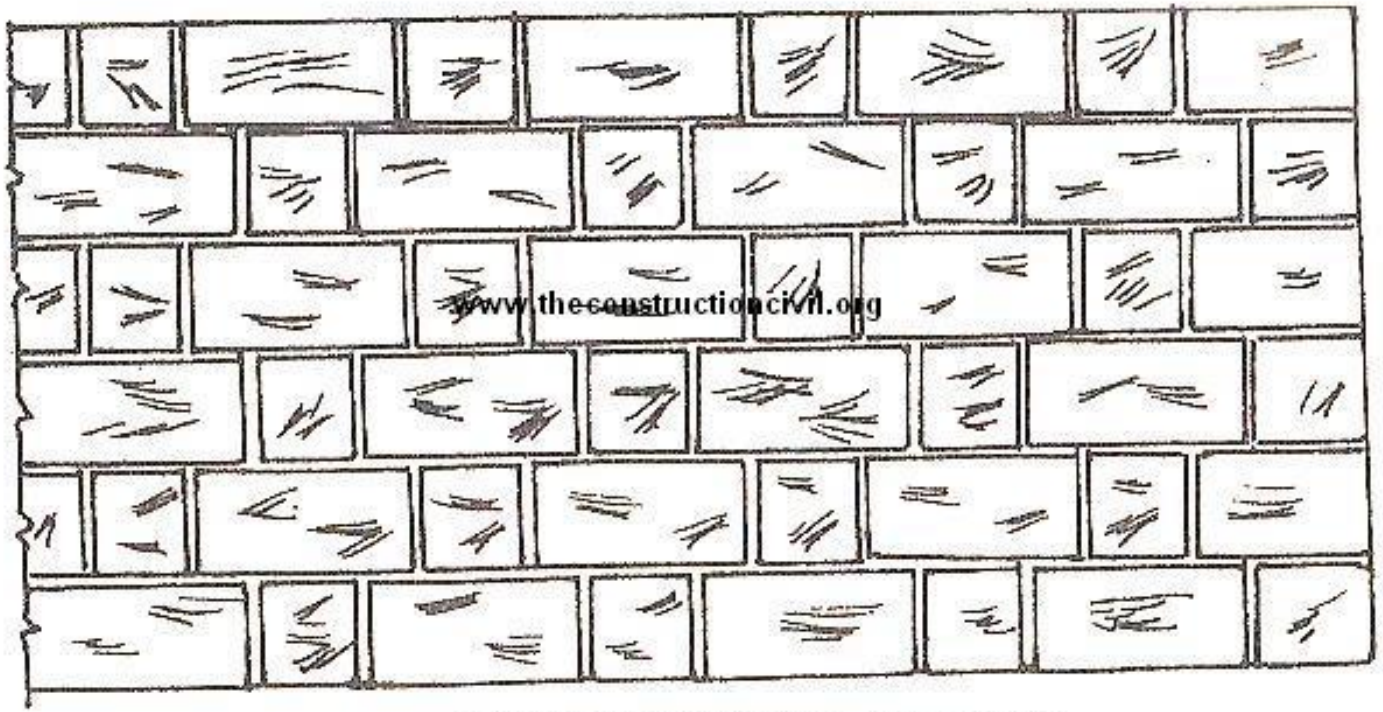
ELEVATION



SECTION

(ii) BUILT TO COURSES RANDOM RUBBLE MASONRY

- In this type of masonry, the work is roughly leveled up to form courses, varying from 12 to 18 in thick.
- In each course, headers of one course height are placed at certain intervals.
- This type of masonry is used to construct residential buildings, go downs, boundary walls etc.



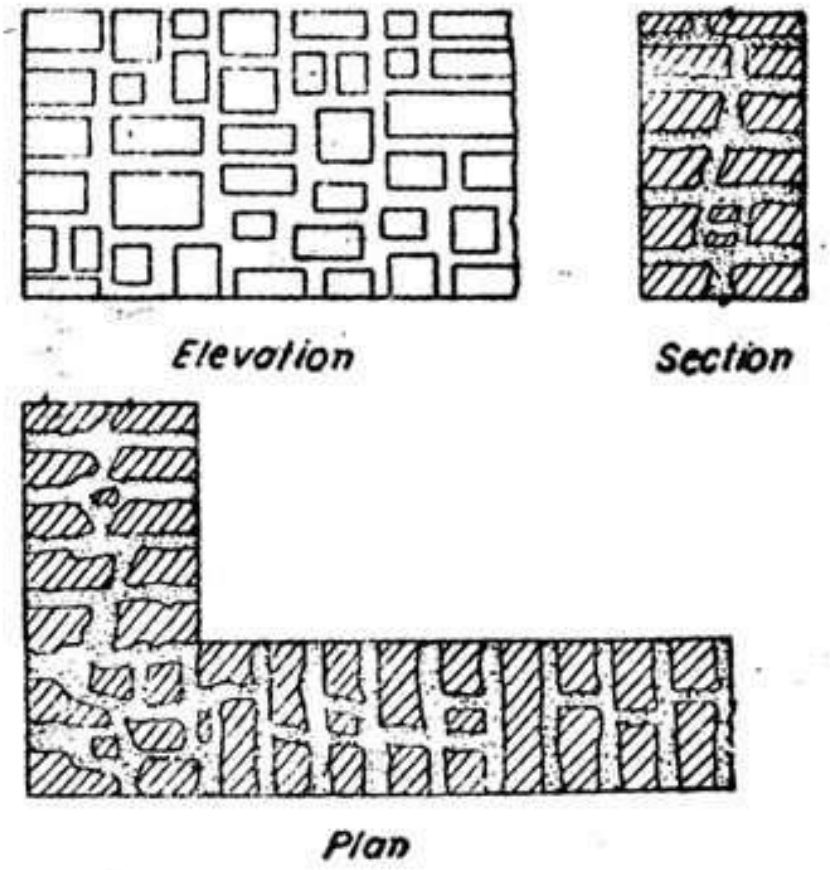
Coursed Rubble Masonry

(b) SQUARED RUBBLE MASONRY

- The Rubble masonry in which the face stones are roughly squared by hammer dressing or chisel dressing, before their actual laying is called "**Squared Rubble masonry**".
- Types of squared rubble masonry:

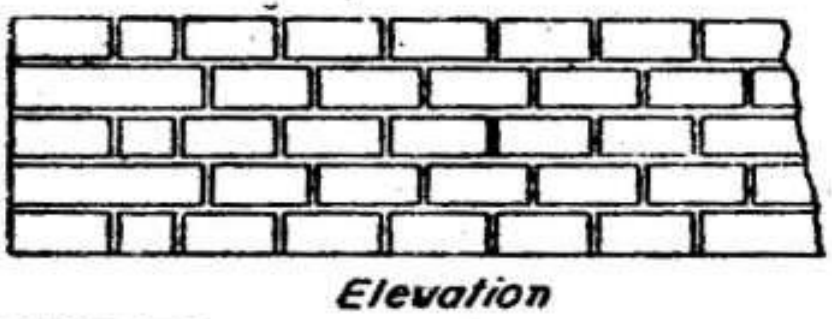
(i) UNCOURSED SQUARED RUBBLE MASONRY:

- In this type of squared rubble masonry, the stones are roughly squared and built without continuous horizontal courses .
- It is used for ordinary buildings in hilly areas, where a good variety of stones are easily and cheaply available



(ii) BUILT TO COURSES SQUARED RUBBLE MASONRY:

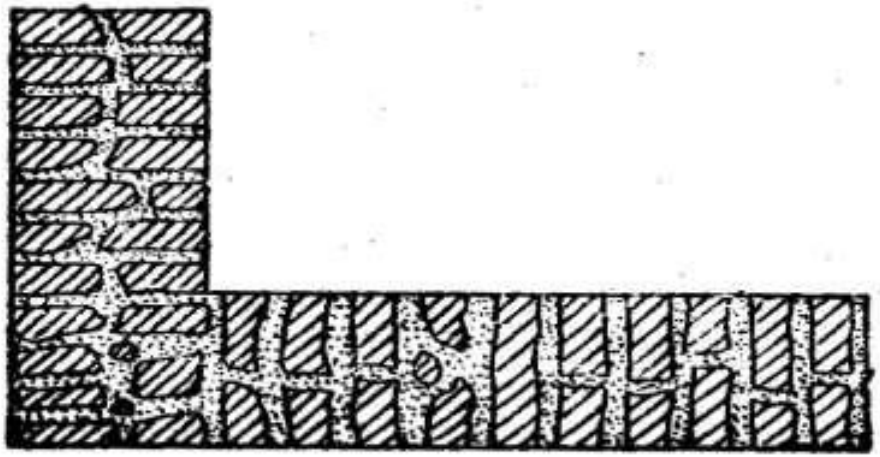
- In this type of stone masonry the uniform height stones are used in horizontal layers not less than 13cm in height.
- Generally, the stone beds are hammered or chisel dressed to a depth of at least 10cm from the face.
- The stones are arranged in such a manner so that the vertical joints of two consecutive curse do not coincide with each other



Elevation



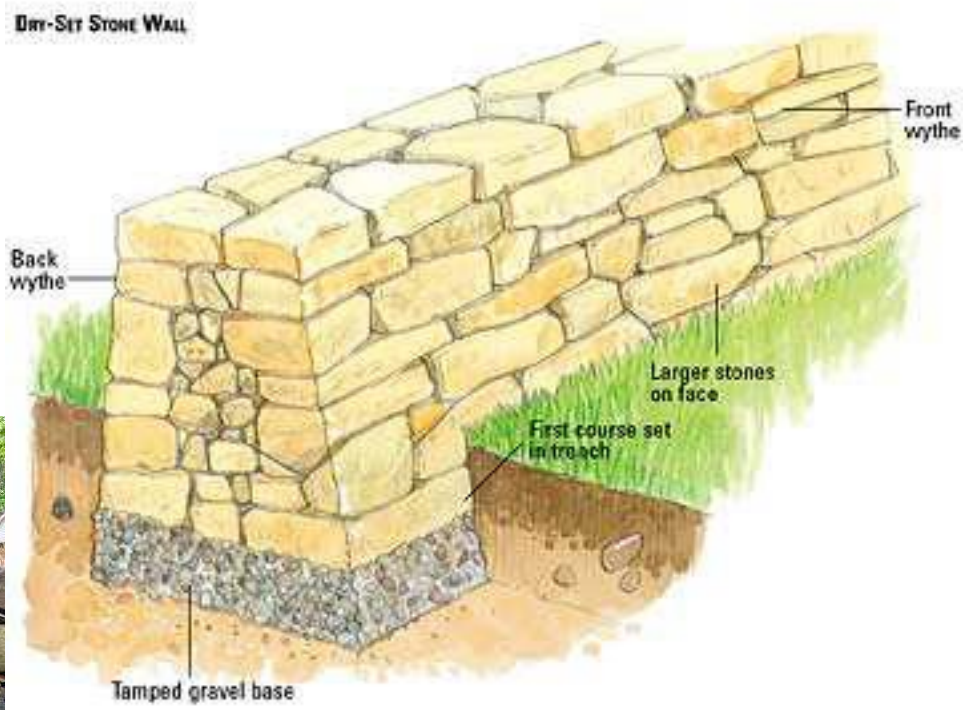
Section



Plan

(c) DRY RUBBLE MASONRY

- The rubble masonry in which stones are laid without using any mortar is known as " **Dry Rubble Masonry".**
- It is an ordinary masonry and is recommended for constructing walls of height not more than 18 ft.
- In case, the height is more, three adjacent courses are laid in Squared rubble masonry, in mortar at 9 ft interval.

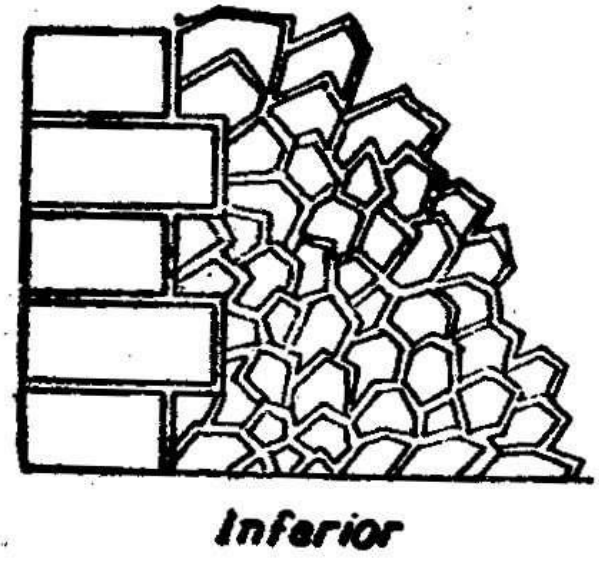


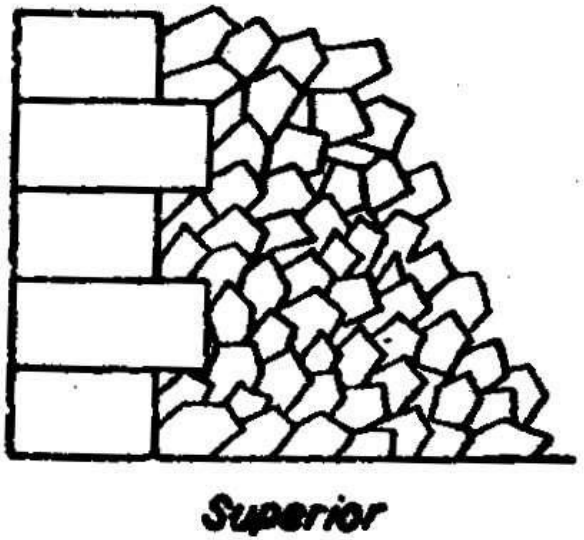


d)Polygonal rubble masonry:

- In this type of masonry the stones are roughly dressed to an irregular polygonal shape.
- The stones should be so arranged as to avoid long vertical joints in face work and to break joints as much as possible.
- Small stone chips should not be used to support the stones on the facing.







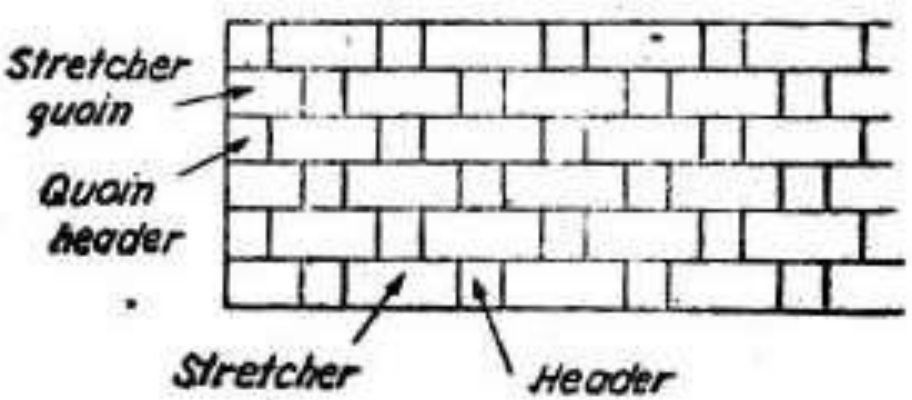
(2) ASHLAR MASONRY

- The stone masonry in which finely dressed stones are laid in cement or lime mortar, is known as "**Ashlar Masonry**".
- In this masonry all the joints are regular, thin, and of uniform thickness.
- This type of masonry is costly in construction as involves heavy cost of dressing of stones.
- This masonry is used for heavy structures, arches, architectural buildings, high piers, abutments of bridges etc.

TYPES OF ASHLAR MASONRY

(a) ASHLER FINE / COURSED ASHLAR MASONRY:

- In this type of masonry stone blocks of same height are used in each course.
- This is finest type of stone masonry work.
- Every stone is fine tooled on all sides so that stone gives perfectly hori. And vertical joints.
- Thickness of Mortar joint is less than 1/16 in and is uniform through out.
- The thickness of mortar joint is kept uniform throughout and it should not be more than 5mm.The exposed joints are finely pointed.



Ashlar Fine Masonry



(b) RANDOM COURSED ASHLAR MASONRY



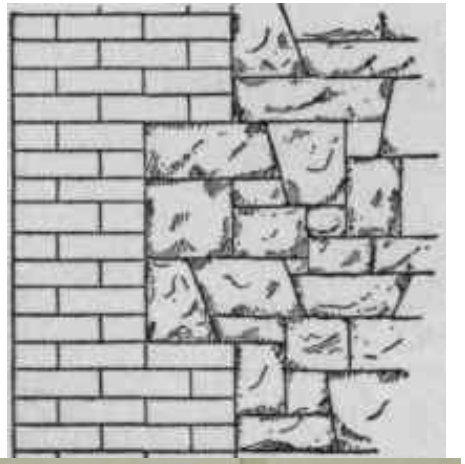
This type of Ashlar masonry consists of fine or coursed Ashlar masonry but the courses are of varying thickness, depending upon the character of the building.

(c) ROUGH TOOLED ASHLER MASONRY



- In this type of masonry, the beds and sides are finely chisel-dressed but the face is made rough by means of tools .
- A strip, about 25mm wide and made by means of chisel is provided around the perimeter of the rough dressed face of each stoneIn this type Ashlar masonry, the sides of the stones are rough tooled and dressed with chisels.
- Thickness of joints does not exceed ¼ in.

(d) QUARRY FACED ASHLAR MASONRY:

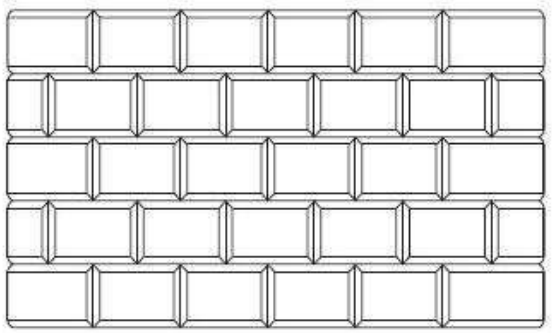


- In this type of masonry, the exposed face of stone is not dressed but is kept as such so as to give rock facing. This type of Construction gives massive appearance.
- The Ht. of each block may vary from 15 cm to 30 cm.
- The thickness of mortar joint may be up to 10mm.

(e) CHAMFERED ASHLAR MASONRY:



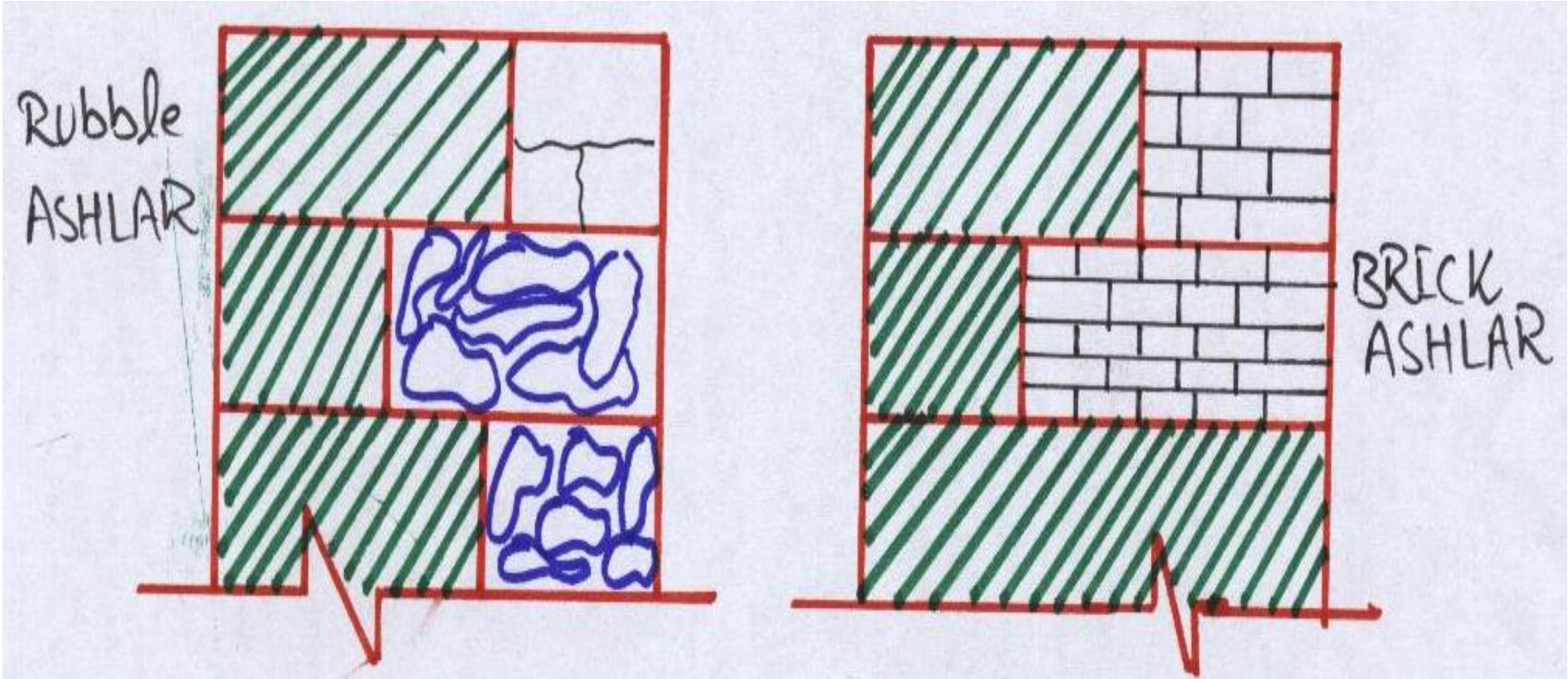
- In this type of masonry, the beds and sides are finely chisel-dressed but the face is made rough by means of tools .
- A strip, about 25mm wide and made by means of chisel is provided around the perimeter of the rough dressed face of each stone but is chamfered at an angle of 45degrees.
- It is similar to Quarry faced except that the edges are given a slope of 45 degree for a depth of 1 in. or more.



(f) ASHLAR FACING

- In order to reduce the cost and to give the appearance of Ashlar facing to the wall it is usual practice to construct walls with facing of Ashlar and backing of Rubble or brick masonry. Such walls are also called "**Compound or Composite masonry walls**".

- If the backing is of Rubble masonry, It is called "**Rubble Ashlar**" and if the backing is of brick work the masonry is termed as "**Brick Ashlar**"



COMPARISON BETWEEN BRICK MASONRY AND STONE MASONRY

(1) **Stone is stronger** and more durable than brick and for public buildings; it is decidedly more suitable than brick. It reflects strength in every inch of it. It is in tune with nature. Its color improves and looks more serene with age.

On the other hand, brick is an artificial product made as a copy of stone. It is flimsy material and plastering is only a camouflage for its defects.

COMPARISON BETWEEN BRICK MASONRY AND STONE MASONRY

(2) **Stone is water proof**. On the other hand, Brick absorbs moisture and with dampness certain salts rise in the walls from the ground and cause disintegration of bricks.

Especially brick should not be allowed to come in contact with urine or sewage and in such places it must always be covered with cement plaster or any other protective coat.

COMPARISON BETWEEN BRICK MASONRY AND STONE MASONRY

(3) **Brick offers greater facility for ornamental work in plaster** as a rough shape can first be given to it by means of any tool. This is not so in case of stones.

(4) **Plaster does not stick so well to stones** as it does to brick.

(5) On account of the regular shape and uniform size of brick, a **proper bond can be obtained with comparative ease.**

COMPARISON BETWEEN BRICK MASONRY AND STONE MASONRY

(6) Due to the **handy size of brick**, brick masonry can be more rapidly constructed than stone masonry.

(7) **Brick wall requires a fixed quantity of mortar** and even with careless masons, the regular shape of the brick considerably reduces the possibility of hollows being left in the body of the wall. This is not so with some stone walls.

COMPARISON BETWEEN BRICK MASONRY AND STONE MASONRY

(8) **It is possible to build brick walls of any thickness** e.g., 4 1/2 in, 3 in, 9 in, 13 1/2 inch etc. Whereas, the minimum thickness of ordinary stone wall is 15 in. Stone walls of a smaller thickness than 15 in, have to be constructed with properly dressed stones, which involves a comparatively high cost.

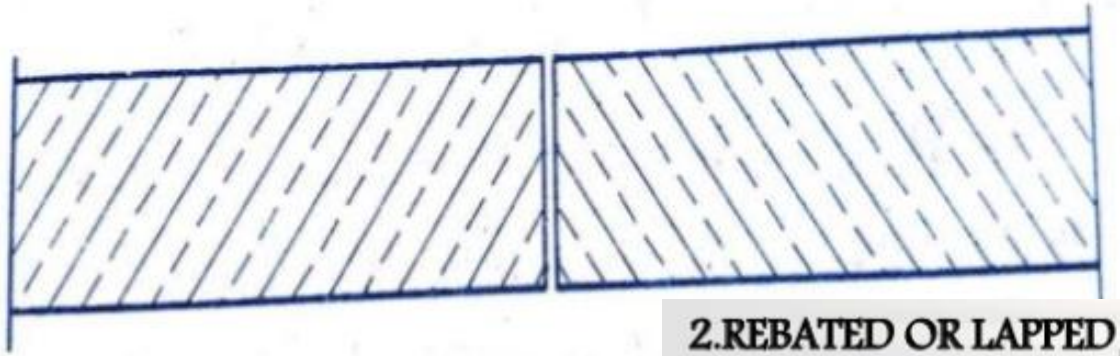
(9) **Brick does not absorb as much heat as stone does**. So, brick is more fire resistant than stone.

Properties	Stone Masonry	Brick Masonry
Strength	High Strength	Less strength
Durability	Excellent	Less
Appearance	No external treatment required	Plastering required
Mortar Joints	Thick	Thin
Danger from dampness	No	Yes
Fire Resistance	Less	More
Handling	Requiring Lifting Devices	Easy to handle
Method of Construction	Quarrying, Dressing	Preparation of Clay, Moulding, Drying, Burning
Cost	More	Less
Use	Pier, Dam, Residential	Residential & Public

JOINTS PRESENT IN STONE MASONARY

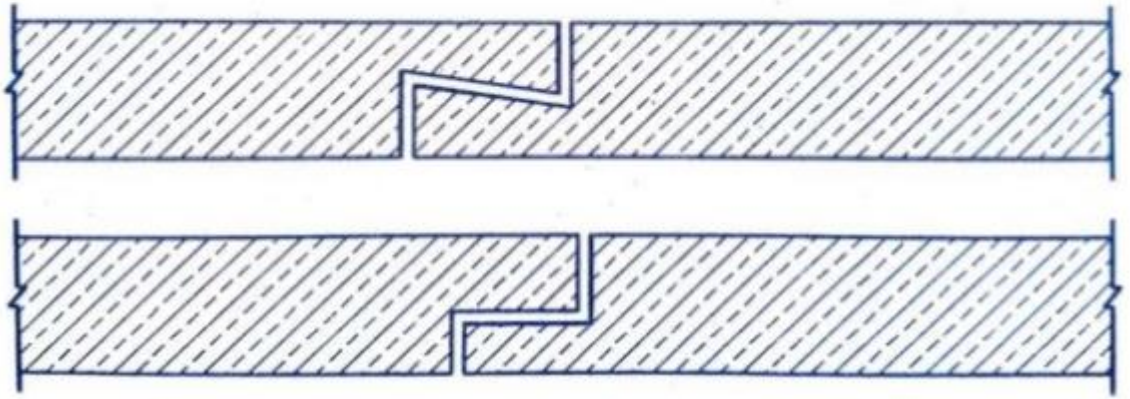
1.BUTT JOINT:–

Dressed edges of two stones are placed side by side.



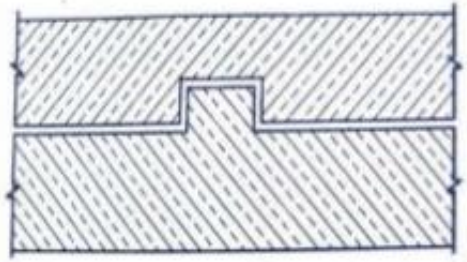
2.REBATED OR LAPPED JOINT:–

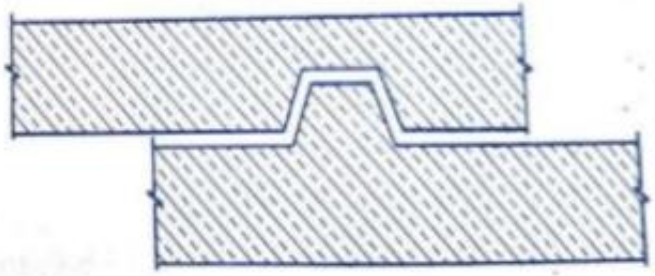
Provided in arches, gables etc. Used in slopes to prevent the movement of stones.

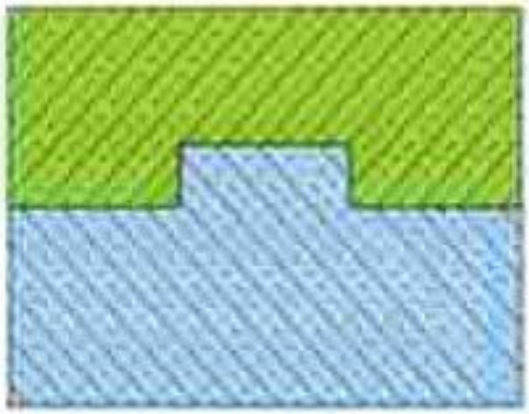


3.TABLE OR BED JOINT:-

Provided in sea wall to prevent the lateral movement of stones. Ht of projection is 30-40 mm & width is 1/3 w of stone.

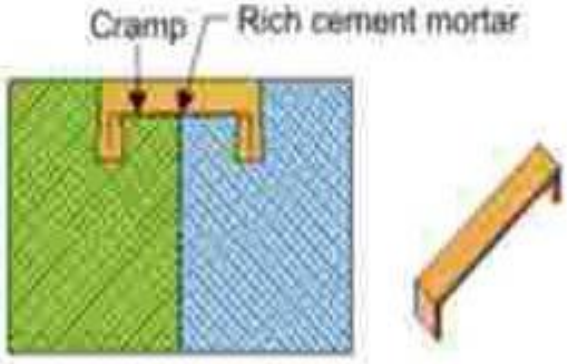


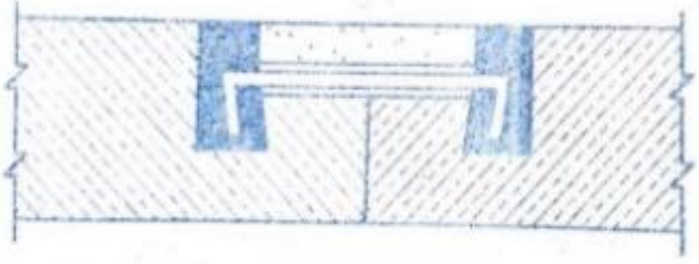


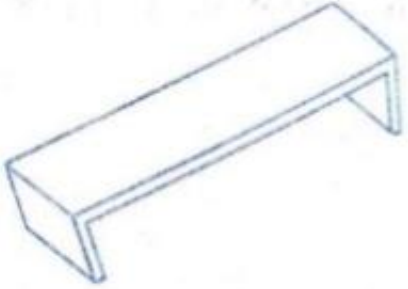


4.CRAMPED JOINT:-

The metal cramps are used to prevent the tendency of the joint to be pulled apart. The dovetail shape holes are made in both the stones to be joined. After placing the cramp in position, the joint is grouted and covered with cement, lead or asphalt.

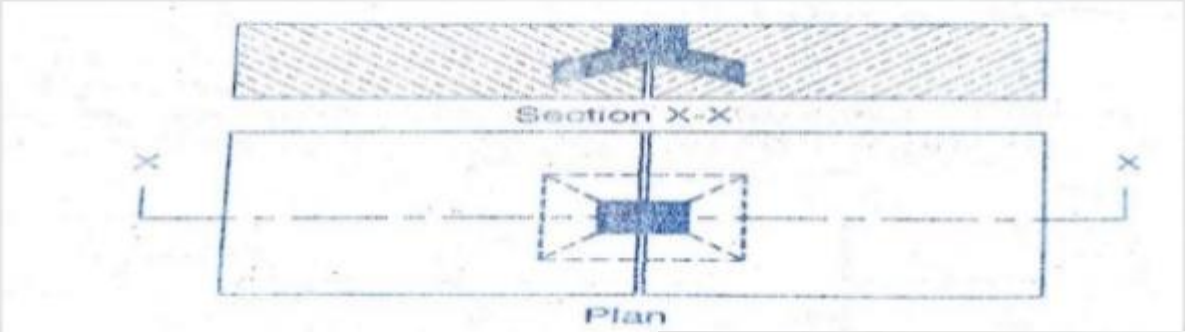


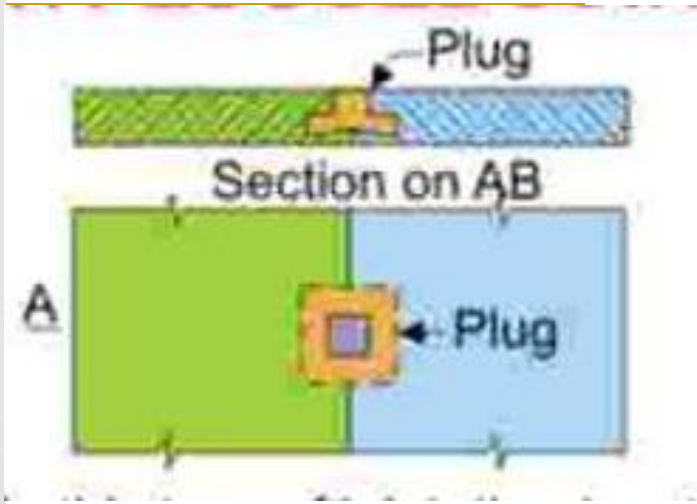


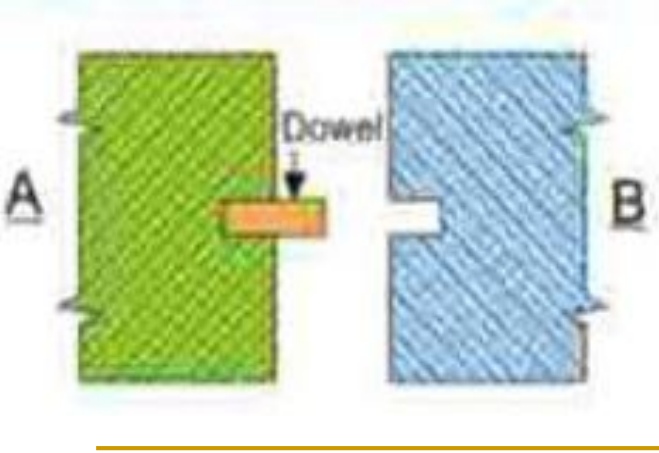


5.PLUG JOINT:-

The plug joint is an Alternative of cramped joint. It consist of plug holes of dovetail shape in plan and cut below the top surface of stones. The stones are jointed by filling the holes with cement.

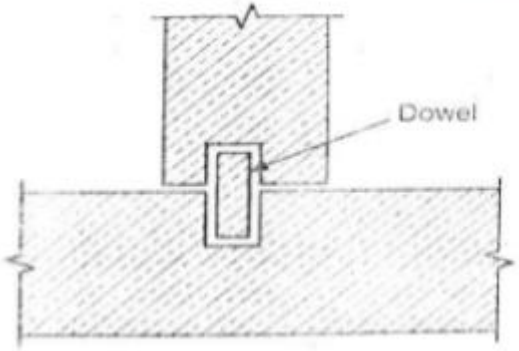


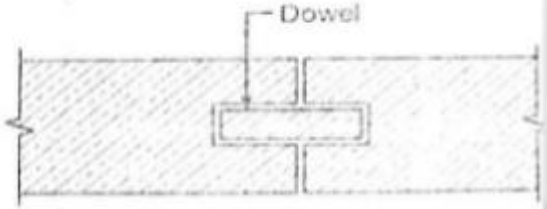




6.DOWEL JOINT:-

This type of joint is used to ensure the Stability of stones against sliding or displacement. Dowels made of slate, copper, brass . The dowels are set in cement mortar.





1. BUTT JOINT



In this type of joint, the square surface of one stone is placed against that of another as shown in figure.

2. REBATED JOINT



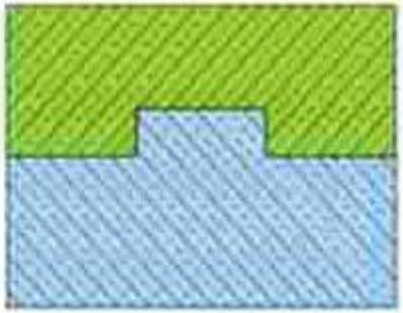
In this type of joint, the rebates are provided which prevent the movements of stones.

3. JOGGLE JOINT



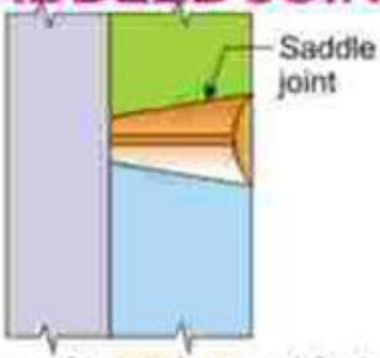
In this type of joint, a projection is kept on one stone and a corresponding sinking is provided in the other stone as shown in figure.

4. TABLED JOINT



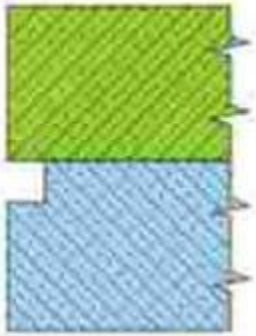
In this type of joint, a joggle is formed in the bed of the stone to prevent lateral movement.

5. SADDLED JOINT

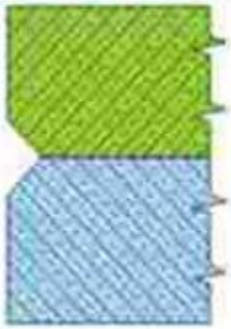


This type of joint, is provided to protect the joints of the cornices and such other weathered surfaces.

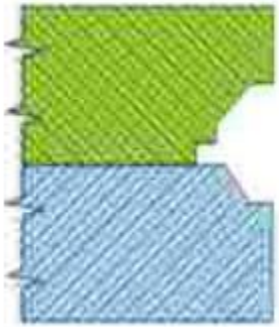
6. RUSTICATED JOINT



Channelled joint



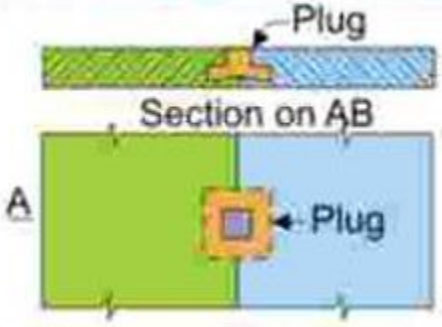
Vee joint



Vee and channelled joint

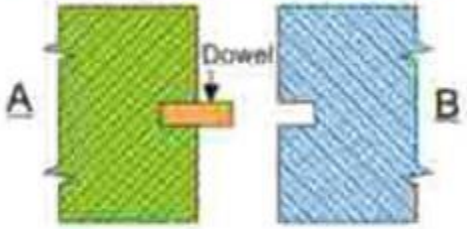
The three forms of rusticated joints are shown in figures. The channelled joint is more common. In this joint, the sinking is made on the lower joint so as to avoid the possibility of entry of water through the mortar joints.

7. PLUGGED JOINT



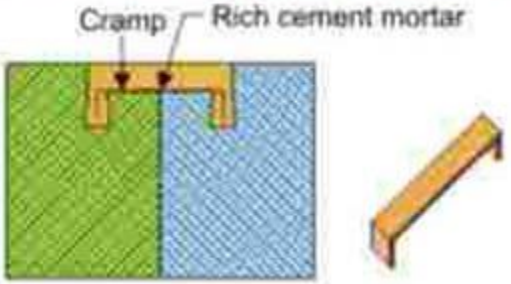
In this type of joint, the dovetail shaped mortices are provided in the sides of adjacent stones as shown in figure.

8.DOWELLED JOINT



The dowelled joint can be easily used in place of joggled joints.

9.CRAMPED JOINT



In this type of joint, the cramps are used instead of dowels.

IMPORTANT POINTS TO BE OBSERVED IN THE CONSTRUCTION OF STONE MASONRY

Stones should be well seasoned, hard, tough, uniform in texture, free from defects (cracks, cavities, patches of loose/soft materials,soil)

Stones should be laid on their natural beds.

Proper bond should be maintained

Vertical joints should be staggered

Vertical faces should be truly in plumb and should be checked by plumb bob.

Stones should be wetted before use

Masonry should be cured for at least two weeks

SELECTION OF STONE FOR STONE MASONRY OR PRINCIPLE FOR STONE MASONRY

Availability of stone-

the stone used for masonry should be easily available from nearby sources.

Ease of working-

the stone should permit easy dressing and shaping to facilitate working on it easily

Appearance-

the appearance of the stone should attractive and colour should be uniform

Strength-

the crushing strength of stone should permit the use of stone in wall masonry

Polishing characteristics-

the stone when polished should look attractive with shining texture.

Composite masonry: It is the one which is constructed out of two or more types of building units or of different types of building materials.

The composite masonry may be adopted due to two reasons:

1. **Improvement in the appearance of walls etc.**
2. **Use of available materials, to obtain optimum economy**.

Composite masonry may be of following types:

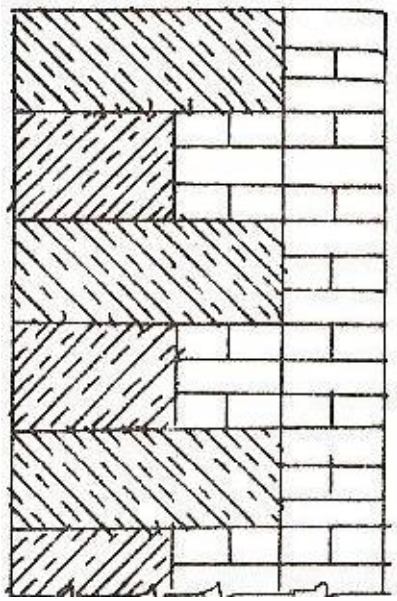
1. Stone Composite masonry
2. Brick Stone Composite masonry
3. Cement Concrete Masonry
4. Hollow clay tile Masonry
5. Reinforced brick masonry

1.Stone Composite Masonry:

It consist of combination of ashlar masonry and rubble masonry.

Rubble masonry is generally very cheap, while ashlar masonry gives pleasing appearance.

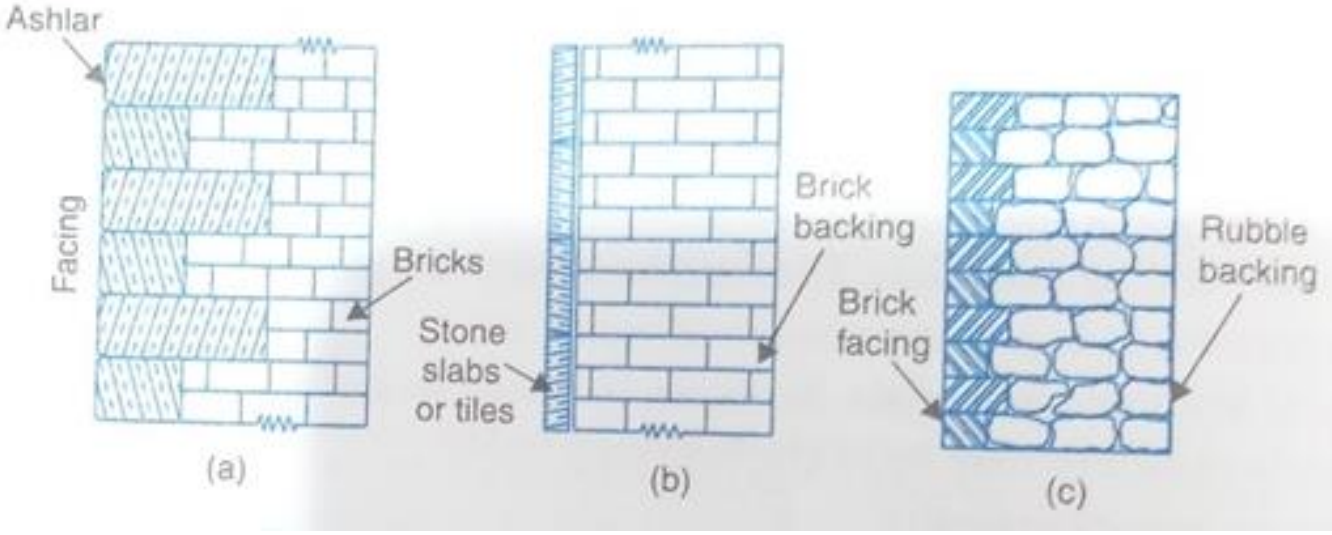
Hence rubble masonry is used in backing of wall while ashlar masonry is used in facing.



2.Brick Stone Composite Masonry:

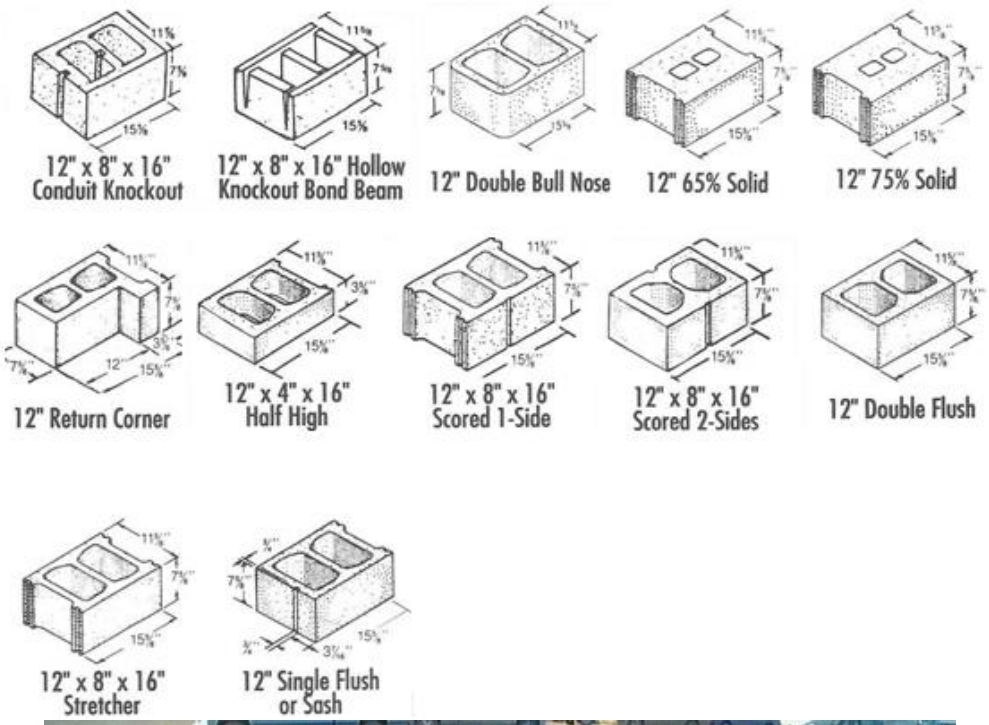
Brick and stone can be simultaneously used in three forms of composite masonry:

i. Brick backed ashlar masonry
ii. Brick backed stone slab facing
iii. Rubble backed brick masonry



3.Concrete Masonry:

- Concrete masonry uses cement concrete blocks, either solid, for wall construction ,with or without stone facing.
- A hollow unit is defined as the unit which has core void area greater than 25% of gross area.
- Various types of concrete units, depending upon size and shape, are manufactured and these can be grouped as
- Regular concrete block: Are manufactured from dense aggregate and used in load bearing walls.
- Hollow concrete block: are manufactured from light weight aggregates. They are used as load bearing as well as non load bearing walls.





4.Hollow Clay Block Masonry:

- This are made of selected clay ,which is dried and burned. The clay blocks are used to build fondations,walls,partitions,floors and other structural members.
- Even though the walls of blocks are relatively thin ,they are quite strong and light.
- This tile are fire proof, resistant to termite and free from decay caused by the contact of moisture or chemicals.



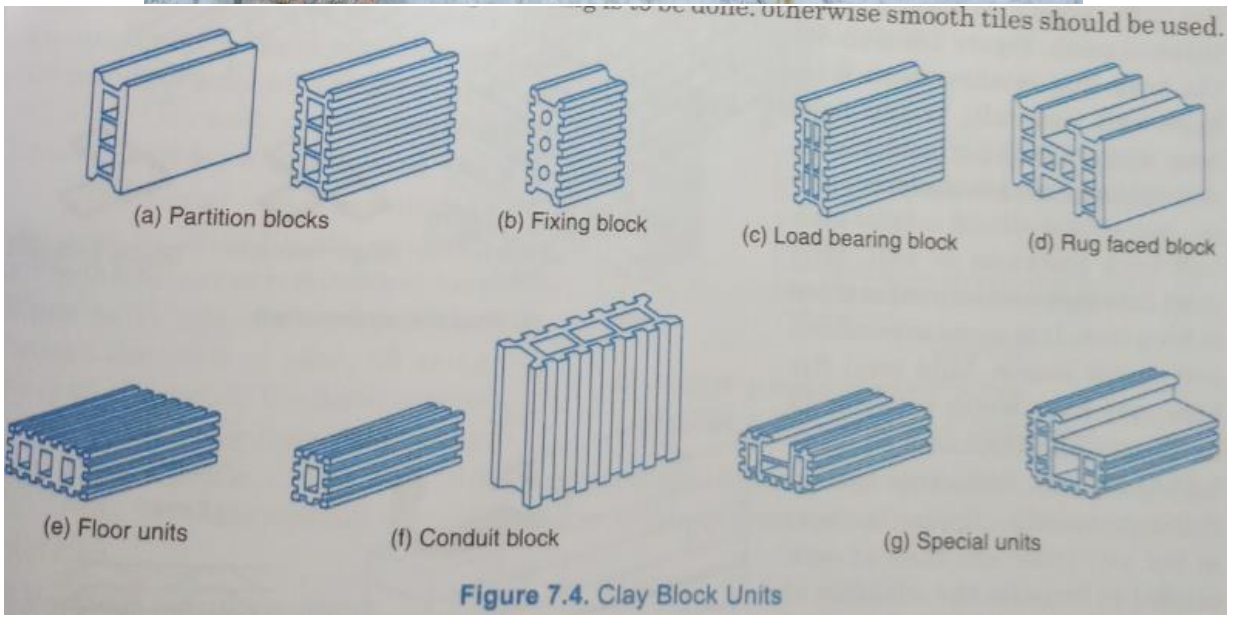
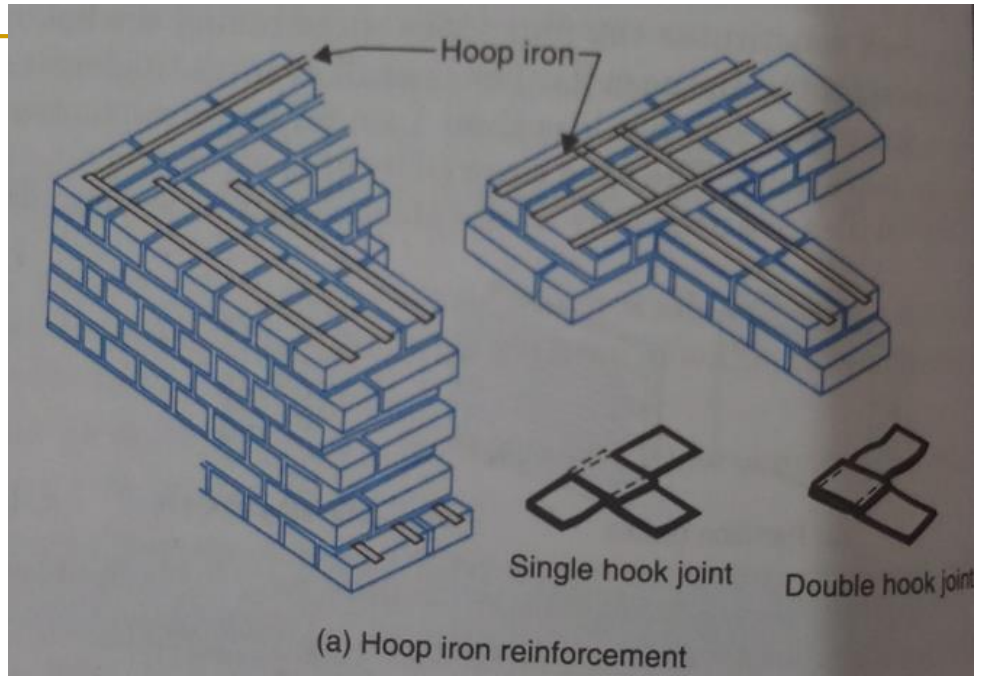


Figure 7.4. Clay Block Units

5.Reinforced brick masonry:

R/f brick work is one in which brick masonry is strengthened by the provision of mild steel falts,hoop iron, expanded mesh or bars. It is adopted in following circumstances:

1. When brick work has to bear tensile and shear stress.
2. When it is required to increase the longitudinal bond
3. When brick work is supported on soil which is susceptible to large settlement.
4. When the brick work is supposed to act as beam or liver over opening.
5. When brick work is to carry heavy compressive loads.



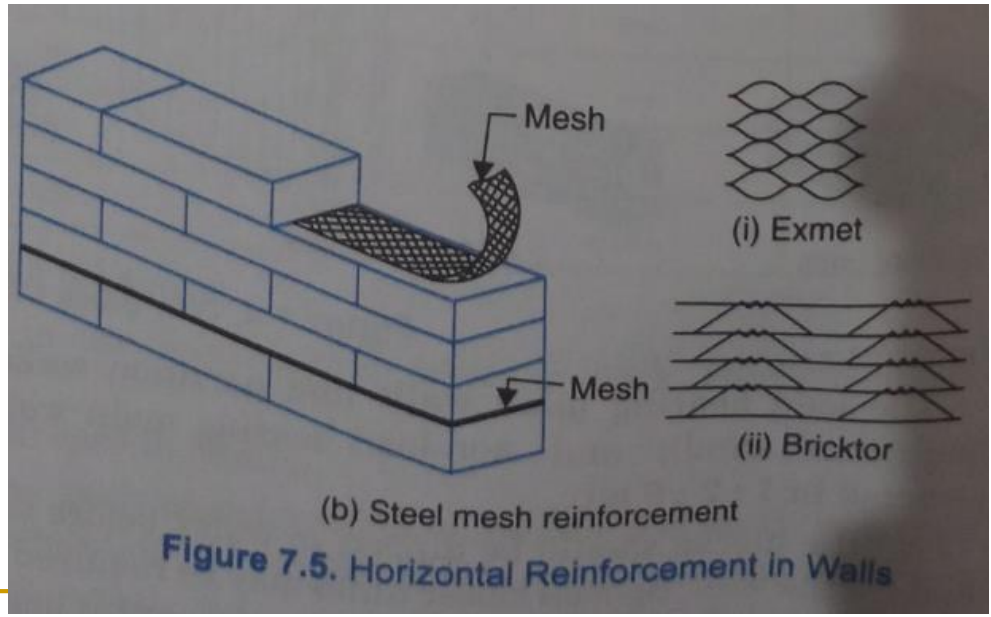


Figure 7.5. Horizontal Reinforcement in Walls

PARTITION WALL

1. It is thin internal wall which is constructed to divide the space within the rooms or areas. A partition wall may be either non load bearing or load bearing.
2. Generally partition walls are non load bearing. A load bearing wall partition wall is called internal wall.
3. The thickness of partition will depend on amount of usable floor space Partition walls are designed as **non-load bearing** walls. It may be of **folding**, **collapsible** or **fixed** type.

- The thickness of partitions will reduce the usable floor space in the building.
- The partition may be taken upto full floor height or upto eye-level i.e. 2000 to 2500 mm.

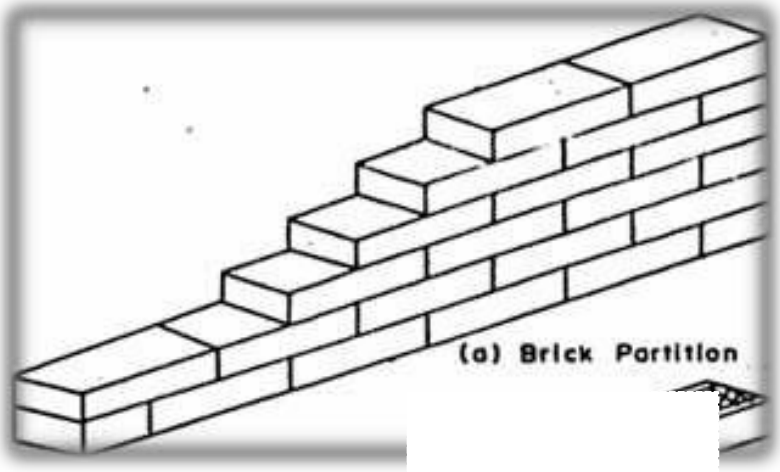
REQUIREMENTS OF PARTITON WALLS

- Should be **strong enough** to carry its own weight.
- Should be strong **capable of resisting impact** developed due to the use of building.
- Should **act as a sound barrier** especially when it divides a room into two units.
- Should possess enough strength **to support some wall fixtures, sinks, wash basins**, etc.
- Should **have the capacity to support suitable decorative surface**.
- Should be **light, thin, cheap, fire-resistant and easy to construct**.
- Shouldffer sufficient **resistance against heat, dampness**, white ant or fungus etc.
- Should **have thin cross-section** so that maximum floor area can be utilized.
- **Provide adequate privacy** in rooms both in respect of sight and sound.
- Should possess proper coherence with the type of building structure.
- **Rigid enough to take the vibrations** caused due to loads.

TYPES OF PARTITION WALLS

- Brick partitions
- Clay block partitions
- Concrete partitons
- Glass partitons
- Timber partitions
- Metal partitions
- Plaster slab partitions
- Asbestos cement sheet partitions
- Wood wool slab partitions
- Strawboard partitions

1.BRICK PARTITIONS

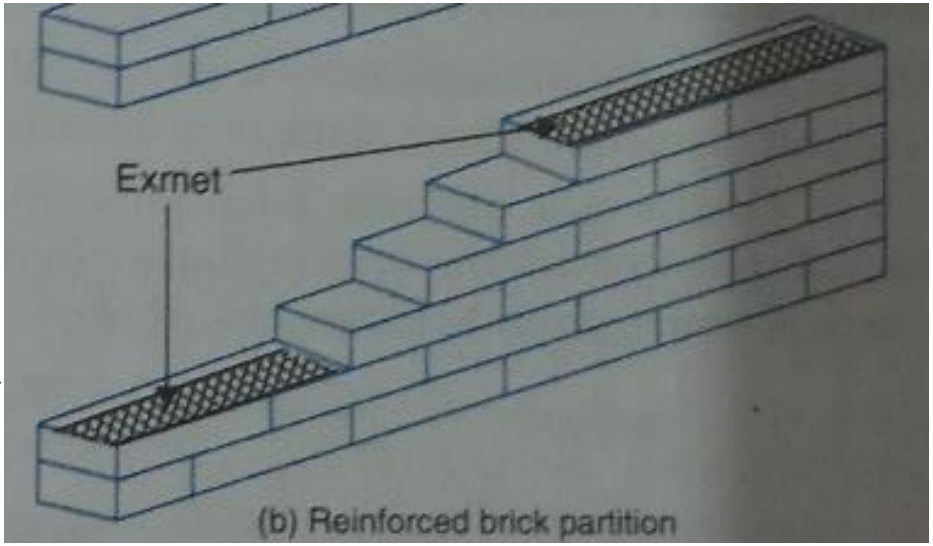


- *PLAIN BRICK PARTITON*
  - **INSTALLATION-** Through normal brick wall construction (stretcher bond).
  - **FEATURES-**
    - Can not take heavy load and their ht.is restricted to about 2000 mm or so.
    - Cheapest and simplest type of partition.
    - Can work as a visual barrier.
    - Upto an extent it can also serve as a sound and heat barrier.



1.BRICK PARTITIONS

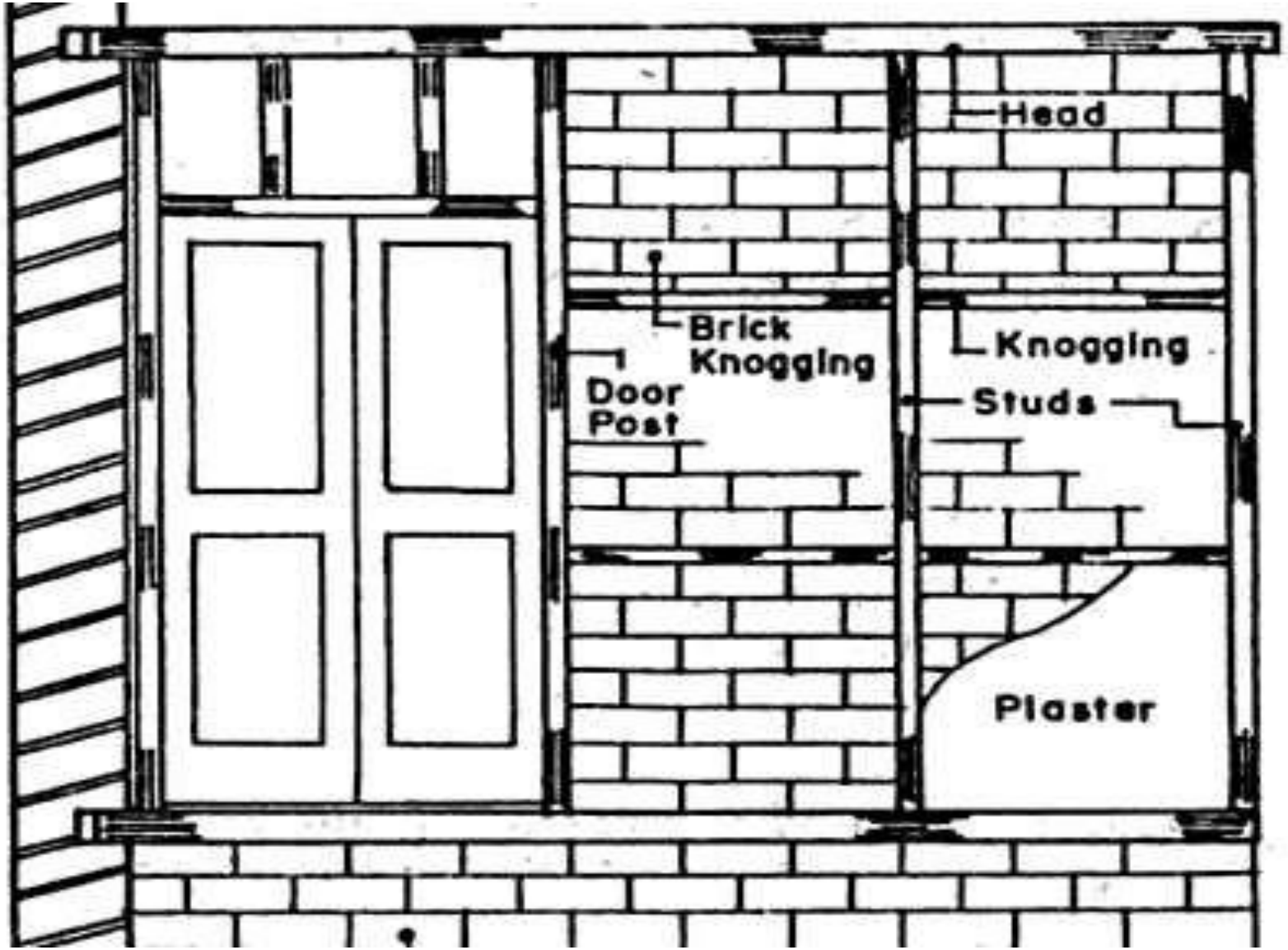
- *REINFORCEMENT BRICK PARTITION*
  - **INSTALLATION-** Through normal RBW construction; the **reinforcement is put up at every third or fourth brick course** and the reinforcement may be in the form of wire mesh or exmet (steel meshed strips) or hoop-iron or steel bars.
  - **FEATURES-**
    a. More durable and possess more strength than the plain one.
    b. Can be of full floor height or restricted to 2000-2500mm
    c. Serves as a visual, sound, fire and heat barrier.



1.BRICK PARTITIONS

- *BRICK NOGGING PARTITION*
  - **INSTALLATION-**
    - Consist of a framework of timber within which 1/2 brick partition are fitted.
    - Framework of timber consists of head, sill, vertical members (studs), horizontal members (noggings), crossed braces and puncheons (studs over door frame).
    - Studs (100x50 mm), head and sill (100x75 mm)
    - The studs are placed at a distance of 3 to 4 times of brick length.
    - The nogging pieces are placed at a vertical distance of about 600-900 mm.
    - The crossed braces are provided at intersections.

1.BRICK PARTITIONS

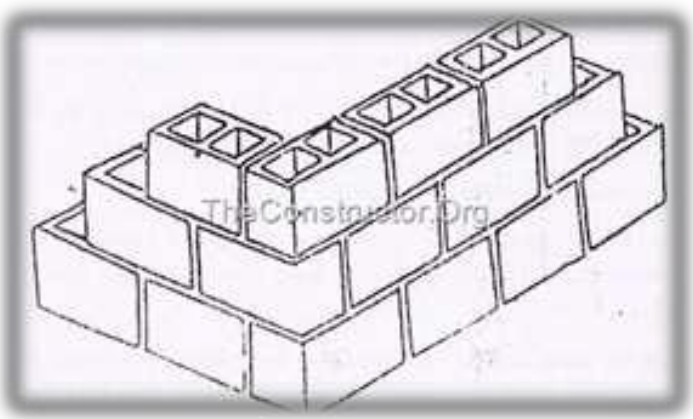


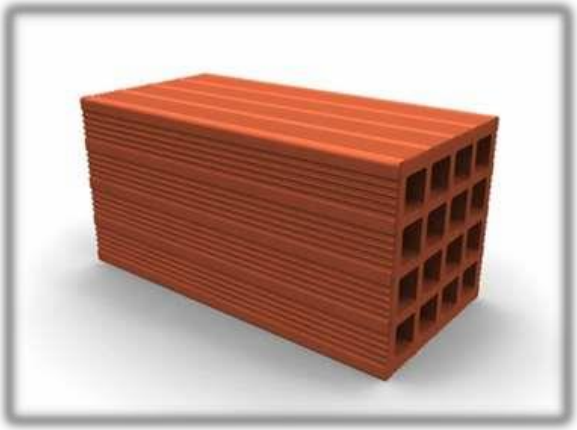
2.CLAY BLOCK PARTITONS

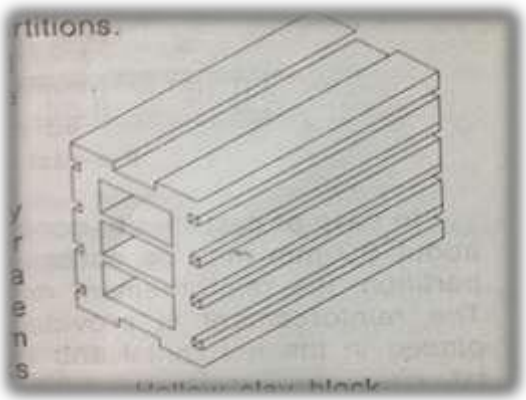
- **INSTALLATION-**
  - prepared from clay or terra-cotta and maybe either solid (300 x 200 x 50-150mm) or hollow (300 x 200 x 400mm).
  - grooves are provided on top, bottom and sides and work as keys to the plaster
  - Mud or normal cement mortar is used as binding material.

- **FEATURES-**
  - Light in weight and are non-shrinkable
  - Serves as a visual, sound, fire and heat barrier.
  - Weaker than the brick partiton in terms of strength.

2.CLAY BLOCK PARTITONS











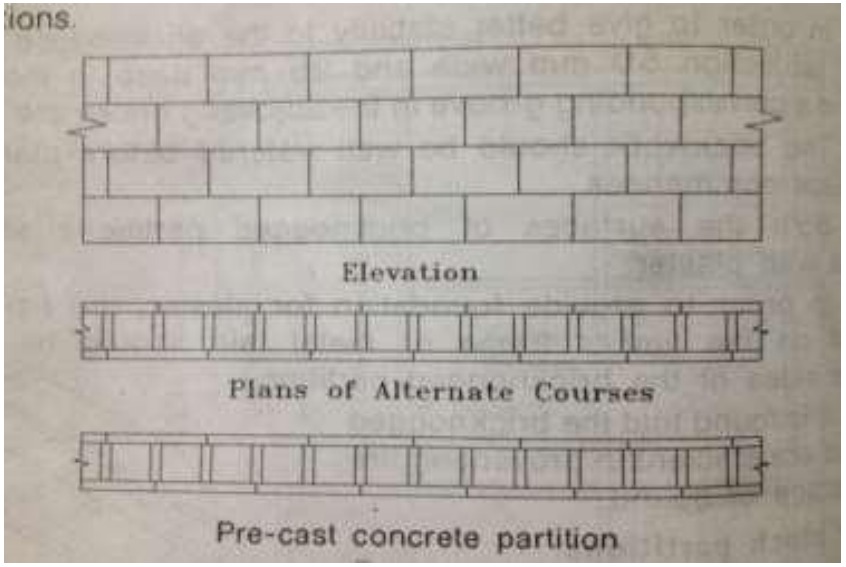
3.CONCRETE PARTITIONS

- *PRE CAST CONCRETE PARTITION*
  - **INSTALLATION-**
    - Concrete(1:2:4) slabs of suitable sizes are prepared and secured to the pre-casr posts.
    - alternatively, the T-shaped or L-shaped units maybe prepared and fixed.
    - thickness is taken about 40mm.
    - cement mortar is used as a binding material.
  - **FEATURES-**
    - Light in weight and are non-shrinkable
    - Serves as a visual, sound, fire and heat barrier.
    - Stronger than the brick and clay block partition in strength wise.

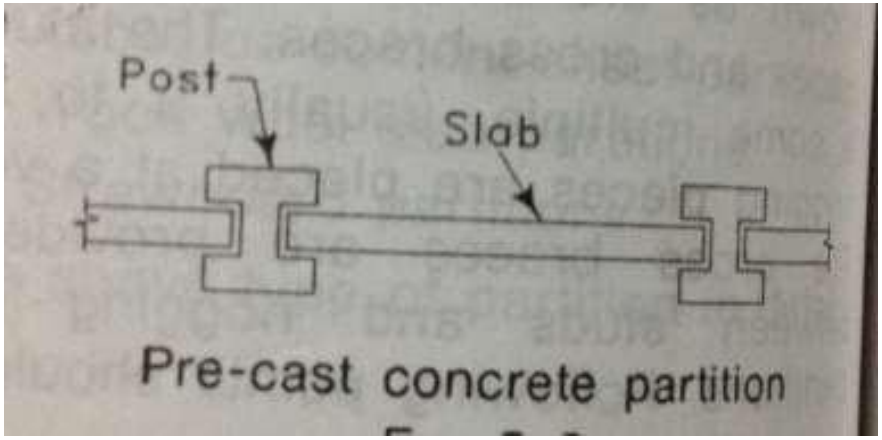
3.CONCRETE PARTITIONS

- *CAST-in-situ CONCRETE PARTITION*
  - **INSTALLATION-**
    - A concrete(1:2:4) wall of thickness about 80mm may be constructed
    - To **get more strength reinforcement may be provided at the centre of thickness** and is placed in the horizontal and vertical directions.
    - cement mortar is used as binding material.
  - **FEATURES-**
    - Light in weight and are non-shrinkable
    - Serves as a visual, sound, fire and heat barrier.
    - Stronger than the brick and clay block partition in strength wise.

3.CONCRETE PARTITIONS













4.GLASS PARTITIONS

- *GLASS SHEET PARTITIONS*
  - **INSTALLATION-**
    - **A timber framework is prepared and then sheets of glass are inserted in the panels**.
    - white cement or a mixture of linseed oil and whiting chalk named putty to kept glass sheets in position.
    - To get more strength reinforcement may be provided at the centre of thickness and is placed in the horizontal and vertical directions.
    - **White cement mortar is used as binding material**.
  - **FEATURES-**
    - Light in weight, vermin- proof, sound- proof and damp proof.
    - Require less space, provide decorative effect through various colors and designs.
    - Delicate in maintainence and can be transparent or can not be.
    - Used at those places where visual barrier is less required.
    - Can be easily cleaned hence more hygenic.

3.GLASS PARTITIONS

GLASS BLOCK PARTITIONS

- **INSTALLATION-**
  - **Do not require any timber framework** upto a maximum panel area of 11m².
  - Keys and grooves are provided for fixing in which white cement mortar is filled.
  - Sometimes ties and rubber beading is also provide for forming rigid joints.
  - White cement is used as a binding material.
- **FEATURES-**
  - Can carry their own self weight only
  - Light in weight, vermin- proof, sound- proof and damp proof.
  - Provide decorative effect through various colors and designs.
  - Delicate in maintainence and can be easily cleaned, hence hygenic.
  - Suitable for those places where window openings cannot be provided and where clear glass is objectionable for the sake of privacy.

4.GLASS PARTITIONS









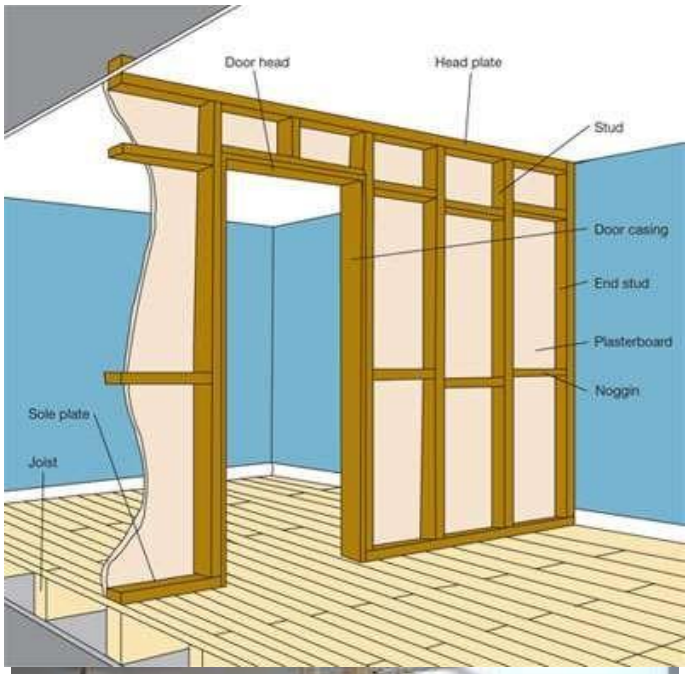




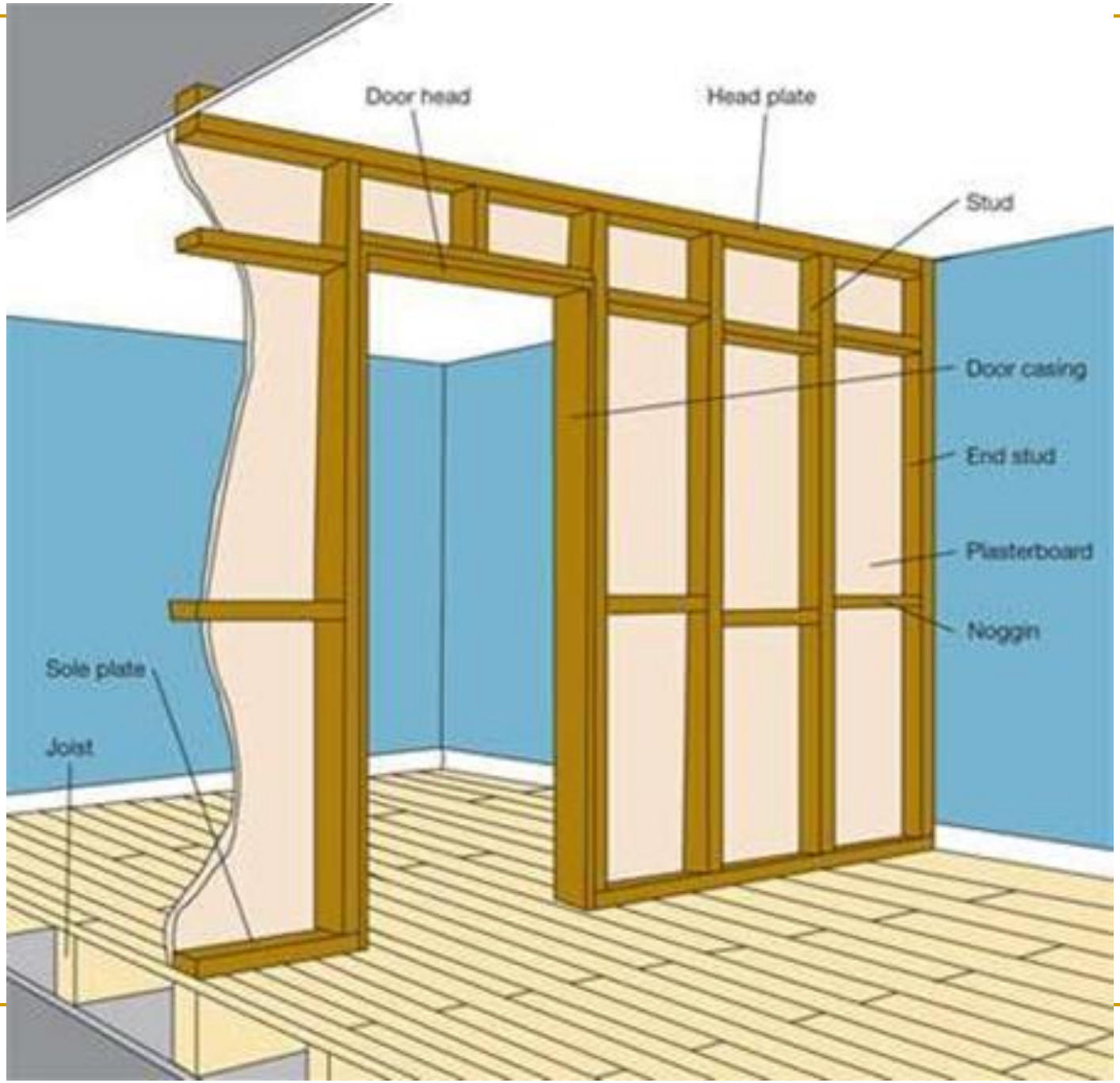
5.TIMBER PARTITIONS

COMMON OR STUD PARTITION

- **INSTALLATION-**
  - The terminologies of wooden framework is same as in **brick nogging partitions**.
  - Solid support is provided for the full length of partition, therefore the sill is placed either at right angles or over the joists of the floor below.
  - The nogging pieces are cut tightly and fixed between the studs and nailed or may be in the form of wooden strips 50x20 mm which can be inserted in the studs and nailed
  - The partition can be finished by inseting wooden panels, etc. in the hollow spaces.






Door head
Head plate
Stud
Door casing
End stud
Plasterboard
Noggin
Sole plate
Joist

5.TIMBER PARTITIONS

- **FEATURES-**
  - Light in weight and therefore can be easily constructed where a base for partition is not available.
  - Transfer some load of wall (rigid and triangulation of framing)
  - Easy to construct.
  - Should not be constructed at basement floors and ground floors normally, due to dampness.
  - Do not possess fire- resisting properties.
  - Not properly insulate the sound ans resist to termites.
  - Can work as a visual barrier.

5.TIMBER PARTITIONS

- *TRUSSED AND BRACED PARTITION*
  - **INSTALLATION-**
    - Similar to the common partitions**. But braces, steel straps and bolts are used in addition to studs, head, sill and nogging pieces**.
    - Behave on the theory of truss, hence they are useful when it is possible to provide supports only at the ends of partiton.
    - To make framework more rigid, a horizontal member, ***intertie*** maybe provided between the head and sill.
    - The arrangement of various members of the framework and their sizes depend on number of openings in the framework and the amount of floor load to be carried by the trussed partition.

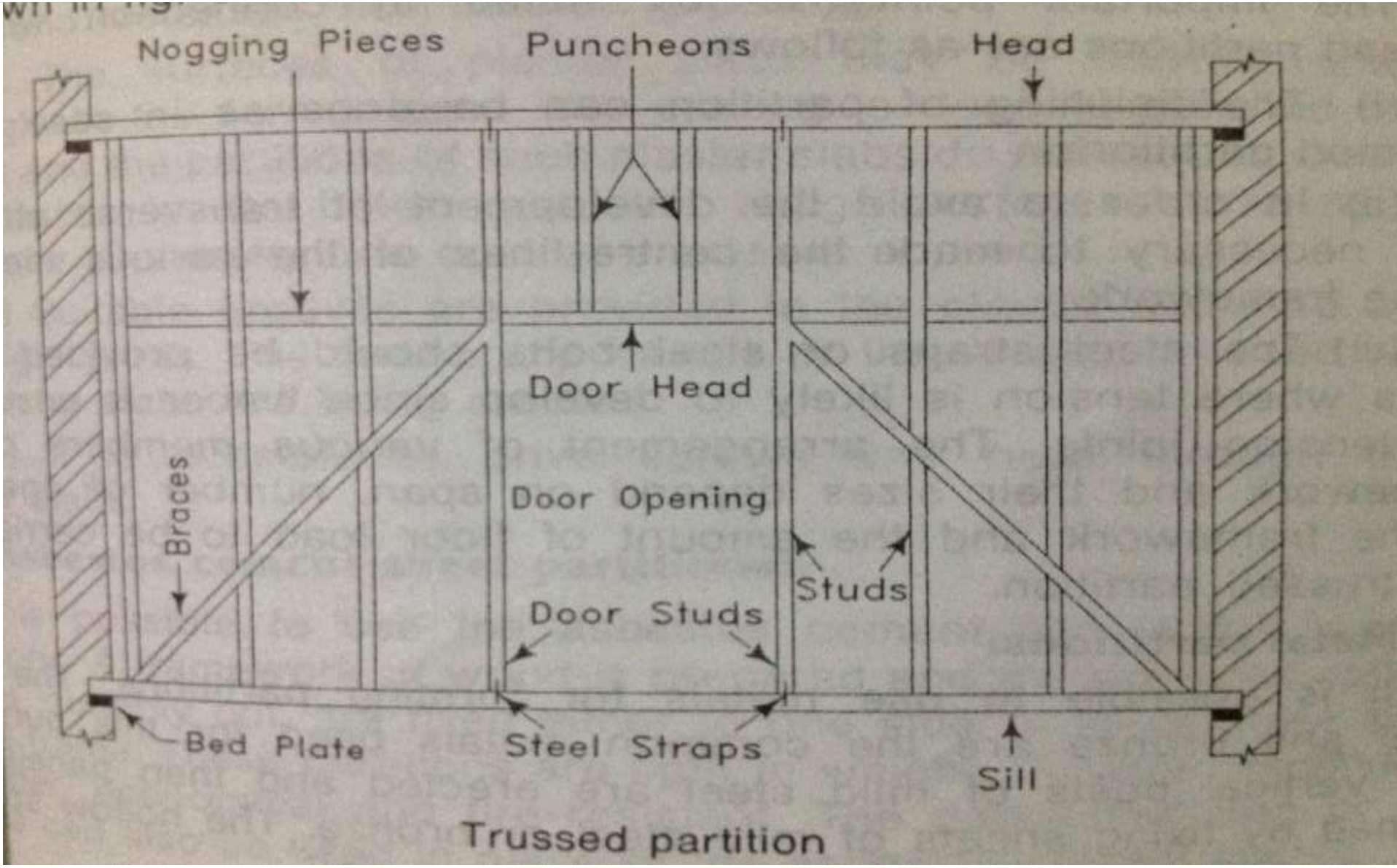




5.TIMBER PARTITIONS

- **FEATURES-**
  - Light in weight and therefore can be easily constructed where a base for partition is not available.
  - Transfer some load of wall (rigid and triangulation of framing)
  - Easy to construct.
  - Should not be constructed at basement floors and ground floors normally, due to dampness.
  - Do not possess fire- resisting properties.
  - Not properly insulate the sound and resist to termites.
  - Can work as a visual barrier.

5.TIMBER PARTITIONS



Trussed partition

6.METAL PARTITIONS

- *Mild steel and bronze* are the common metals used for this purpose

- **INSTALLATION-**
  - Vertical posts of mild steel are erected and then panels are formed by fixing sheets of mild steel or bronze.
  - The hollow spaces may be filled with some good insulating material.
  - When metal laths are used they are fixed by wires with 9mm or 12mm diameter steel rods or steel channels.
  - The expanded metal lath is commonly used for this purpose.
  - Plastering is done on both the sides.
  - The thickness of completed partition is about 50mm.

6.METAL PARTITIONS

- **FEATURES-**
  - Light in weight, vermin- proof and damp proof.
  - Easy to construct and easy to remove.
  - Suitable for office buildings, industrial buildings.Should not be constructed at basement floors and ground floors normally, due to dampness.
  - Possess fire- resisting properties.
  - Not properly insulate the sound.
  - Can work as a visual barrier.

6.METAL PARTITIONS







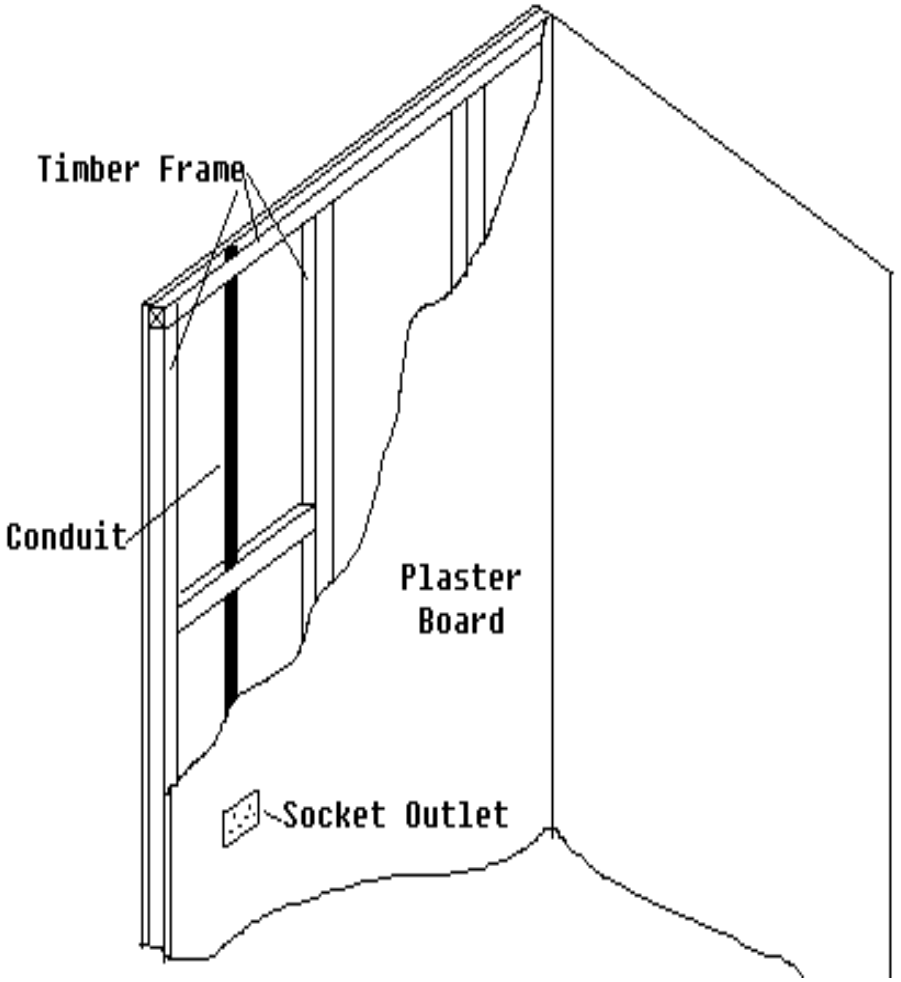




7.PLASTER SLAB PARTITONS

- Generally made of burnt *gypsum* or *plaster of paris* and sawdust is added in it to reduce its density.
- Prepared in moulds of iron or wood.
- **INSTALLATION-**
  - Plaster boards (1000-2000 x 300 x 50-100mm) either solid or hollow are provided with grooves so as to form rigid joints.
  - The temporary frames or liners may be used when the plaster slabs are being erected.
  - Screws and nails can also drive in these plaster slabs.
- **FEATURES-**
  - Light in weight and are non-shrinkable
  - Serves as a visual, sound, fire and heat barrier.

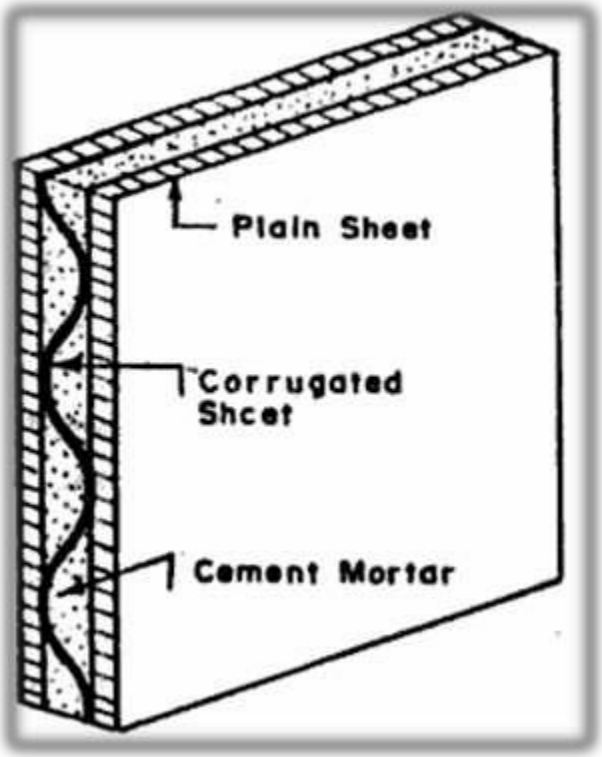
7.PLASTER SLAB PARTITONS





8.ASBESTOS CEMENT SHEET PARTITONS

- **INSTALLATION-**
  - A framework of wood is prepared and asbestos sheets are fixed either to one side or to both the sides of the frame.
  - Absestos cement slab partition generally consist of two sheets of 10 mm thickness and they are attached to an inner corrugated sheet (5mm).
  - The slabs are placed in position and joined by cement mortar and surface is then painted or distempered.



- **FEATURES-**
  - Light in weight, durable, impervious to water and economical.
  - Serves as a visual, sound, fire and heat barrier.

9.WOOD WOOL SLAB PARTITIONS

- Prepared from a mixture of *portland cement* and *wood wool* or *wood shavings*. A small quantity of *Gypsum* is sometimes added.
- **INSTALLATION-**
  - The slabs are placed in position and joined by cement mortar and surface is then painted or distempered.
- **FEATURES-**
  - Light in weight, durable and economical.
  - Serves as a visual, sound, fire and heat barrier.
  - Get effected by moisture easily.







ANY QUESTIONS?